SECURITY LEVEL	CONFIDENTIAL

APP Phone Connection User Guide

EEASYTECH Confidential

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Revision History

Revision No.	Description	Date
1.0	Initial release version.	2021-12-17
1.1	Add the sp field in getproductinfo; Add the curcamid field in Getdeviceattr, indicating which camera is currently being previewed (such as front camera/rear camera); Add the transport field (Transmission protocol of RTSP, such as “tcp” or “udp”.) in getmediainfo; Add file_del message event definition； Add the message event definition of storage card status change; Modify the unit of Unix timestamp to “s”.	2021-12-19
1.2	Add logo_osd (logo watermark switch) in the setting items； Add the interface about obtaining the recording duration; Add park_monitor (parking monitoring switch) in the setting items; Add time_lapse (timelapse recording switch) in the setting items; Distinguish between normal setting items and algorithm setting items, and add the ADAS setting item; Add the BSD setting item; Modify the parameter about the current value of querying recording file duration; Modify the description about RTSP realtime stream.	2022-04-11
1.3	Modify the folder name in the getfilelist from emr/recycle to loop/emr/event/park.	2022-05-18
1.4	Add the interface about paging acquisition file list; Delete the interface about algorithm class; Delete park_monitor and time_lapse; Add the interface about timelapse recording frame rate; Add the interface about parking monitoring recording duration; Add the interface about parking monitoring collision sensitivity.	2022-06-08
1.5	Add the interface about entering menu mode; Add the interface about entering playback mode; Add the interface about parking monitoring mode; Add the interface about parking monitoring switch; Add the file identification notes; Modify the description about the SD card state of the storage information interface; Modify the description about RTSP realtime stream.	2022-08-05
1.6	Add the interface about obtaining device capacity (option).	2023-01-28

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1.7	Add the interface about bit file uploading (option); Add a bit in the interface of obtaining device capacity set (option) for selecting whether to playback the video on the APP through small streaming.	2023-03-09
1.8	Add the interface about RTSP video playback small streaming control (option).	2023-03-24
1.9	Add the interface about time-zone setting; Add the interface about restarting Wi-Fi.	2023-04-24
2.0	Add the interface about GPS status change message.	2023-07-06
2.1	Modify the selection of GPS data service; Add a bit to control whether the APP photo is supported; Add a bit to control whether the Wi-Fi name and Wi-Fi password can be modified by the APP; Add the GPS data file protocol; Add the GPS data to write to the video protocol.	2023-07-07
2.2	Add the otaver and wifireboot fields in the getdeviceattr; Add the interface about file locking; Add the message interface about device file state change; Add the message interface about camera plugging state change; Add the message interface about GPS signal state change.	2023-07-13
2.3	Add the interface about light frequency; Add the interface about rear mirror switch; Add the interface about wide dynamic switch; Add the interface about sound switch; Add the interface about screen standby duration; Add the interface about auto poweroff duration; Add the interface about ADAS switch; Add the interface about boot sound switch; Add the interface about video flip switch; Add the interface about video mirror switch; Add the interface about key tone switch.	2023-07-15
2.4	Adjust document structure.	2023-07-29
2.5	Add the description about pkg file uploading; Add the description about RTSP 8554 port.	2023-11-21

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1 About This Document

1.1 Purpose

This document provides reference information for the programmers that use Wi-Fi and other network applications to connect with APP. The document provides the device interfaces from the optimization of various dimensions such as interface scalability, easy-to-use, time and performance, so as to guide the device to better realize the connection with the APP.

1.2 Related Product

This document applies to the following product platforms:

- n7v1
- n7v3
- n7v7
- n7v9

1.3 Intended Audience

This document is intended for:

- Software development engineers
- Technical support engineers

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2 Function Description

2.1 Interface Protocol Type

The connection between APP and devices is mainly divided into the following aspects:

1. HTTP-based information query, status query, device setting, playback, deletion and download of historical files on the device;
2. Real-time picture playback of streaming media based on RTSP or playback of historical files on the device;
3. The device information report based on TCP persistent connection is used to inform the APP of the device status change caused by non-APP operation, so that the APP can synchronize the status.

2.2 Data Protocol Format

The json protocol is used for the transmission of response messages, which occupies a small space and is easy to parse, and the unified format is as follows:

```
{
    "result" : 0 ,
    "info" : {
        "product" : "recorder",
        "company" : "smarttech"
    }
}
```

Field description:

- result: Result code of the API. 0: The query succeeds; other values: The query fails.
  - 0: Success;
  - Other values: Error code for failure (defined by the device);
  - 99: Unknown exception.
- info
  - If `result` is non-0, it indicates the corresponding error description, such as “the operation is unsupported”.
  - If `result` is 0, it is the return data corresponding to this interface, and the data fields are specifically defined by this command.

2.2.1 HTTP Request

At present, HTTP is mainly operated by GET method, and the parameters of GET request are transferred through the query string in URL. The query string is the part that follows the “?” (question mark in English input) in URL, which can contain multiple parameters. The format of the parameters is “parameter=value”, and each

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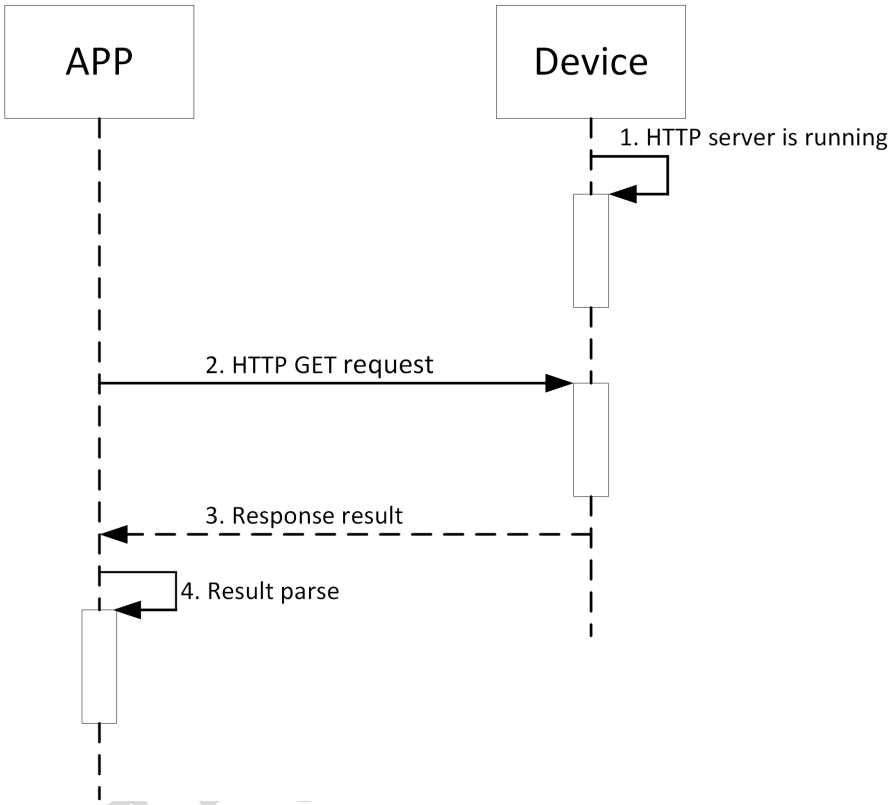
parameter is separated by "&" symbol. For example, the APP dynamically switches to a request to pull back the camera to preview the pushflow: http://192.168.169.1/app/setparamvalue?param=switchcam&value=1.

The running process of HTTP is as follows:

Figure 2-1 Running Process of HTTP



2.2.2 RTSP Real-Time Streaming

- RTSP is a multimedia streaming protocol used to control sound or video, and allows multiple streams to be controlled at the same time, and the network communication protocol used for transmission is not within its defined range, and the server can use UDP or TCP to deliver the streaming content.
- Support streaming media protocol of RTSP for viewing the real-time image. The RTSP implements **RTSP over UDP** or **RTSP over TCP**.
- Real-time images have a high demand for the delay and fluency of images. Generally, the delay is required to be less than 500 ms, and it is best to reach about 200 ms from end to end. It is necessary to control the loss of good delay at both the device end and the APP end.
- The bit rate is required to be less than 2 Mbps, which can ensure the fluency and low delay of images.

Specific implementation process is as follows:

1. The client initiates an RTSP OPTION request to get a method provided by the server. The methods provided by RTSP usually include: OPTIONS, DESCRIBE, SETUP, TEARDOWN, PLAY, PAUSE, SCALE, GET_PARAMETER.

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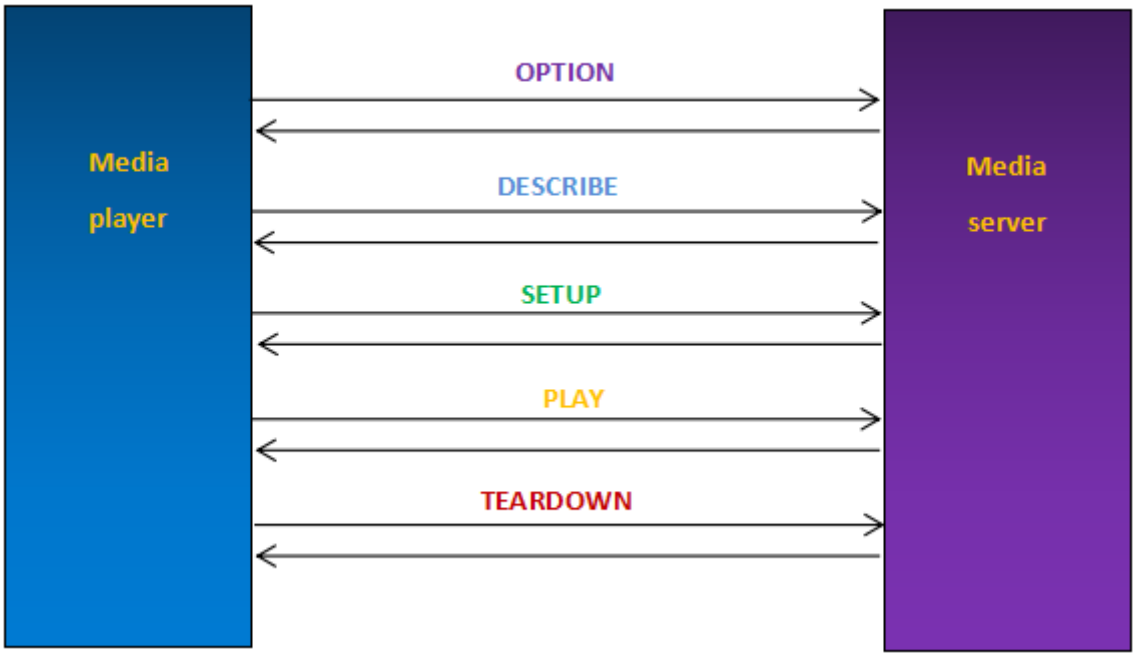
2. The server responds to RTSP OPTION, and the server responds to whatever methods it implements. In this system, only OPTIONS, DESCRIBE, SETUP, PLAY, TEARDOWN methods are implemented.
3. The client initiates an RTSP DESCRIBE request, the server receives information such as the name of the media, the decoding type, and the video resolution, so that obtain the session description from the server (SDP).
4. The server responds to RTSP DESCRIBE, and sends necessary media parameters. When sending H.264, H.265 or AAC, the information mainly includes SPS/PPS, name of the media, and transmission protocol.
5. The client initiates an RTSP SETUP request to request session establishment and ready for transmission. The requested information includes transmission protocol and port index of the client.
6. The server responds to RTSP SETUP, and sends the corresponding port index and session identifier of the server.
7. The client initiates an RTSP PLAY request to playback audio and video streams.
8. The server responds to RTSP PLAY, the responded message includes session identifier, serial number and timestamp of RTP package. At this time, the server packages and transmits the audio and video streams (AAC, H.264 or H.265).
9. The client initiates an RTSP TEARDOWN request to close the connection and stop the transmission.
10. The server closes the connection and stops the transmission after responding the RTSP TEARDOWN request.

The instance implementation process is as follows.

Figure 2-2 RTSP Real-Time Streaming



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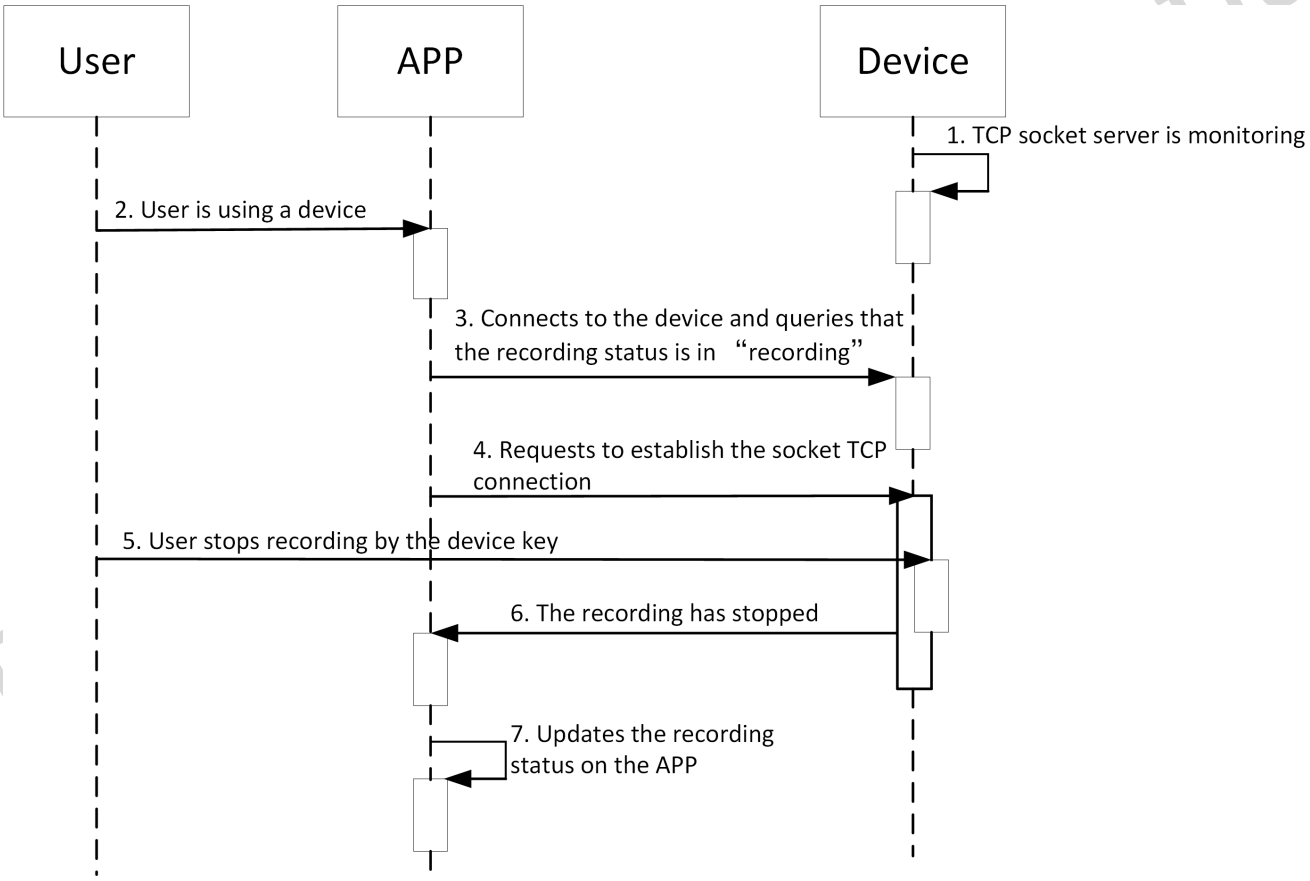
2.2.3 TCP Push Message

The TCP push message indicates the device status change caused by other conditions (not the operation on the APP), which needs to be notified to the APP so that the APP can synchronize the current device state; For example, when the APP is connected to the device, a key of the user on the device changes the recording status of the current device, so the device needs to send such a message to the APP to update the recording status.

The message channel of active messages is a TCP-based socket persistent connection channel, the device is at the server listening end and the APP is at the client end. When the user uses the device on the APP, it will actively establish a connection with the device, and the APP will always obtain the messages sent by the device, so as to synchronize the status changes of the device in time.

The instance process is as follows. Take pushing the recording status change as an example.

Figure 2-3 Pushing the Recording Status Change



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3 Interface Type

Interfaces are classified based on the service, mainly described as follows:

1. Basic information: Inherent properties and important status attributes of the device, such as hardware version and software version of the device;
2. Normal setting of devices: Such as camera streaming switch (front, rear, etc), microphone switch;
3. Device management: Include the modification of Wi-Fi information, the formatting of the SD card, and the restoration of factory settings;
4. Device file management: Obtain file list, obtain file thumbnail, playback file, download file, and delete file.
5. Setting item: Configure the device parameters.

3.1 Querying Class

Obtain inherent properties and current status of a device by information querying class interfaces, such as the query of SD/TF card status information, refer to Obtain Storage Information.

3.2 Setting Class

The interface used to set the setting items of a device usually contains three kinds of contents:

1. Query the parameter list allowed to be set in this setting item. Refer to List of All Parameters for the format.
2. Query the current value of the setting item. Refer to Current Value of Speaker Volume for the format.
3. Modify the current value of the setting item. Refer to Switch the Camera for the format.

Query the parameter list of a setting item (The queried HTTP interface has two methods):

1. When all setting items are queried, the parameter list corresponding to the setting item is parsed.
2. Directly specify interface parameters to query the parameter list of this setting item.

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4 Specific Implement Interface

4.1 Basic Information

4.1.1 Obtain the Product Information

[Interface Function]

Obtain the product model, product manufacturer (brand), chip platform, and solution provider.

[Request Interface on the APP]

http://192.168.169.1/app/getproductinfo

[Request Parameter on the APP]

None.

[Return Parameter on the Device]

Parameter	Type	Description
model	string	Product model, such as “recorder”.
company	string	Product manufacturer, such as “lombotech”.
soc	string	Chip platform, such as “lombotech”.
sp	string	Solution provider, such as “lombotech”.

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "model" : "recorder",
        "company" : "lombotech",
        "soc" : "lombotech",
        "sp" : "lombotech"
    }
}
```

NOTE

If there is no one to fill in above, you can remove the item. For example, if there is no the name of the solution provider, you can remove the ”sp” field.

4.1.2 Obtain the Device Information

[Interface Function]

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Obtain the basic information of the device.

[Request Interface on the APP]

http://192.168.169.1/app/getdeviceattr

[Request Parameter on the APP]

None.

[Return Parameter on the Device]

Parameter	Type	Description
uuid	string	Unique flag of a device. The device can use mac address or SN.
softver	string	Version of the current firmware.
otaver	string	OTA version number that is used to upgrade. It must meet the definition specification of the version number. If the softver meets the specification of OTA version number, this field may not be used.
hwver	string	Hardware version of the current device.
ssid	string	SSID of Wi-Fi.
bssid	string	MAC address of Wi-Fi.
camnum	int	Number of cameras.
curcamid	int	ID of the current camera. 0: front camera; 1: rear camera. If there is only a camera, the ID is 0.
wifireboot	int	Modify Wi-Fi name and Wi-Fi password, whether to restart Wi-Fi with APP after restoring the factory settings (0: not necessary; 1: necessary)

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "uuid" : "112233445566",
        "softver" : "v1.2.3",
        "otaver" : "v1.20230711.1",
        "hwver" : "v1.2",
        "ssid" : "AP-11223344",
        "bssid" : "112233445566",
        "camnum" : 2,
        "curcamid" : 0,
        "wifireboot" : 0
    }
}
```

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4.1.3 Obtain the Storage Information

[Interface Function]

Obtain the storage device (SD/TF card) information of the device: Status of SD/TF card (Normal insertion, unformatted, pulled out, unknown exception) and current available capacity/total capacity.

[Request Interface on the APP]

http://192.168.169.1/app/getsdinfo

[Request Parameter on the APP]

None.

[Return Parameter on the Device]

Parameter	Type	Description
status	int	0: SD/TF card is normal； 1: SD/TF card is unformatted； 2: SD/TF card is pulled out； 3: SD/TF card has damaged； 10: Card is locked and cannot be written； 11: Low speed card, it is suggested to change； 12: SD/TF card is abnormal, it needs to be formatted； 13: Formatting； ... 99: Unknown exception.
free	int	Remaining capacity (in MB).
total	int	Total capacity of SD/TF card (in MB).

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "status" : 1,
        "free" : 12000,
        "total" : 64000
    }
}
```

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4.1.4 Obtain the Battery Status

[Interface Function]

Obtain the information about the battery capacity.

[Request Interface on the APP]

http://192.168.169.1/app/getbatteryinfo

[Request Parameter on the APP]

None.

[Return Parameter on the Device]

Parameter	Type	Description
capacity	int	0-100. Percentage of the current battery capacity.
charge	int	0: Not charging； 1: Charging

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "capacity" : 0,
        "charge" : 1
    }
}
```

4.1.5 Obtain the Media Information

[Interface Function]

Obtain the media information of the device.

[Request Interface on the APP]

http://192.168.169.1/app/getmediainfo

[Request Parameter on the APP]

None.

[Return Parameter on the Device]

Parameter	Type	Description
rtsp	string	It is used to playback the server address of RTSP.
transport	string	Transmission protocol of RTSP, such as “tcp” or “udp”.

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Parameter	Type	Description
port	int	Monitoring port of the TCP socket server on the device of pushing message.

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "rtsp" : "rtsp://192.168.169.1",
        "transport" : "tcp",
        "port" : 5000
    }
}
```

4.1.6 Obtain the Recording Duration

[Interface Function]

Get the current recording duration. It is the recording duration of the current video file, not the continuous recording time. For example, each recording file is 3 minutes, the duration is [0-180s].

[Request Interface on the APP]

http://192.168.169.1/app/getrecduration

[Request Parameter on the APP]

None.

[Return Parameter on the Device]

Parameter	Type	Description
duration	int	Duration of the current recording file (in s).

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "duration" : 56
    }
}
```

4.1.7 Obtain the Device Capacity

[Interface Function]

Obtain the tag of some customizable function on the current device, so as to adapt on the APP.

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[Request Interface on the APP]

http://192.168.169.1/app/capability

[Request Parameter on the APP]

None.

[Return Parameter on the Device]

Parameter	Type	Description
value	string	From left to right. **Zeroth bit**: GPS data serve, where: 0: Not support GPS； 1: Only support GPS data acquisition of downloaded video files; 2: Support GPS data acquisition of downloaded video files and online playback video files; 3: Support GPS data acquisition of downloaded video files, online playback video files and real-time pictures; 4: Support GPS data acquisition of downloaded video files and real-time pictures; **1st bit**: Device type, where: 0: Recorder (by default); 1: Motion camera (The APP can display the battery capacity or charging status of the device after selecting the motion camera); **2nd bit**: Whether to support parking monitoring photo album, where; 0: Supported (by default); 1: Not supported; **3rd bit**: Whether to support file locking on the device through APP operation, where; 0: Not supported (by default); 1: Supported; **4th bit**: Whether to support deleting the locked file, where; 0: Supported (by default); 1: Not supported;

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Parameter	Type	Description
		5th bit: Whether to switch to the photo mode when taking photos, where: 0: Switch the mode to snapshot or use the default mode of the current SoC (by default); 1: Do not switch the mode, directly snapshot the video stream; **6th bit**: Whether to playback the video on the APP through small streaming, where: 0: Playback source file of audio and video file (HTTP Range); 1: Playback small streaming (RTSP over TCP); 2: Playback small streaming (RTSP over UDP); **7th bit**: Whether to support the photo function, where: 0: Supported (by default); 1: Do not support the photo function, only support recording. When some device resources are not enough to support the photo function, the photo button will be hidden on the APP by modifying this field. **8th bit**: Whether to support modifying Wi-Fi name or Wi-Fi password on the APP, where: 0: Support modifying Wi-Fi name or Wi-Fi password (by default); 1: Do not support modifying Wi-Fi name, support modifying Wi-Fi password, the Wi-Fi name will be hidden at this time; 2: Support modifying Wi-Fi name, not support modifying Wi-Fi password, the Wi-Fi password will be hidden at this time; 3: Do not support modifying Wi-Fi name and Wi-Fi password, the Wi-Fi name and Wi-Fi password will be hidden at this time; **nth bit**: To be defined.

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "value" : "011001"
    }
}
```

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4.2 Device Management

4.2.1 Set the Time-Zone

[Interface Function]

Synchronize the time-zone of the mobile phone/other terminal, and when the mobile phone/other terminal is connected to the device, grant the time-zone of the mobile phone/other terminal to the device.

[Request Interface on the APP]

http://192.168.169.1/app/settimezone

[Request Parameter on the APP]

Parameter	Type	Description
timezone	int	Time-zone of the mobile phone/other terminal. For example: ?timezone=8 indicates GMT+8:00; ?timezone=-5 indicates GMT-5:00.

[Return Parameter on the Device]

None.

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : "set success"
}
```

4.2.2 Set the Date and Time

[Interface Function]

Synchronize the data and time of the mobile phone/other terminal, and when the mobile phone/other terminal is connected to the device, grant the date and time of the mobile phone/other terminal to the device.

[Request Interface on the APP]

http://192.168.169.1/app/setsystime

[Request Parameter on the APP]

Parameter	Type	Description
date	long	Year/month/date/hour/minute/second (yyyyMMddHHmmss) of mobile phone/other terminal. For example: ?date=20210123153650

[Return Parameter on the Device]

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None.

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : "set success"
}
```

4.2.3 Modify the Wi-Fi Name

[Interface Function]

Modify the Wi-Fi name of the device.

[Request Interface on the APP]

http://192.168.169.1/app/setwifi

[Request Parameter on the APP]

Parameter	Type	Description
wifissid	String	Modify the Wi-Fi name of the device. For example: ?wifissid=AP-11223344.

[Return Parameter on the Device]

None.

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : "set success"
}
```

4.2.4 Modify the Wi-Fi Password

[Interface Function]

Modify the Wi-Fi password of the device.

[Request Interface on the APP]

http://192.168.169.1/app/setwifi

[Request Parameter on the APP]

Parameter	Type	Description
wifipwd	String	Modify the Wi-Fi password of the device. For example: ?wifipwd=12345678.

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[Return Parameter on the Device]

None.

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : "set success"
}
```

4.2.5 Restart the Wi-Fi

[Interface Function]

Restart the Wi-Fi process after the APP sets the Wi-Fi name and Wi-Fi password.

[Request Interface on the APP]

http://192.168.169.1/app/wifireboot

[Request Parameter on the APP]

None.

[Return Parameter on the Device]

None.

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : "reboot success"
}
```

4.2.6 Format the SD/TF Card

[Interface Function]

Format the SD/TF card of the device. After the formatting is completed, the contents of SD/TF card will be emptied.

[Request Interface on the APP]

http://192.168.169.1/app/sdformat

[Request Parameter on the APP]

None.

[Return Parameter on the Device]

None.

[Example of Device Return after Success]

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```
{
    "result" : 0,
    "info" : "set success"
}
```

4.2.7 Restore the Factory Setting

[Interface Function]

Restore the setting items of the device to the default status of the factory.

[Request Interface on the APP]

http://192.168.169.1/app/reset

[Request Parameter on the APP]

None.

[Return Parameter on the Device]

None.

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : "set success"
}
```

4.2.8 Switch the Camera

[Interface Function]

For devices with multiple cameras, the switch of stream is completed on the device when switching the lens of real-time RTSP stream. If the device wants the switch of stream to be completed on the APP, it is necessary to define the RTSP address separately.

[Request Interface on the APP]

http://192.168.169.1/app/setparamvalue

[Request Parameter on the APP]

Parameter	Type	Description
param	String	Parameter name: switchcam. It is used for lens switch. For example: ?param=switchcam.
value	int	Index of the current lens or logic image. For example: ?param=switchcam&value=1 0: Front;

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Parameter	Type	Description
		1: Rear; 2: Picture-in-Picture

[Return Parameter on the Device]

None.

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : "set success"
}
```

4.2.9 Photo

[Interface Function]

Trigger the photo operation of the device. After the device performs the instruction, the device will generate a photo.

[Request Interface on the APP]

http://192.168.169.1/app/snapshot

[Request Parameter on the APP]

None.

[Return Parameter on the Device]

None.

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : "snapshot success"
}
```

4.2.10 Lock

[Interface Function]

The current normal record is changed to an emergency lock file type.

[Request Interface on the APP]

http://192.168.169.1/app/lockvideo

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[Request Parameter on the APP]

None.

[Return Parameter on the Device]

None.

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : "lock success"
}
```

4.2.11 Enter the Recorder

[Interface Function]

When the device is successfully connected from the connection interface and enters other interfaces, the APP sends this request so that the device knows that the APP is successfully connected.

[Request Interface on the APP]

http://192.168.169.1/app/enterrecorder

[Request Parameter on the APP]

None.

[Return Parameter on the Device]

None.

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : "set success"
}
```

4.2.12 Exit the Recorder

[Interface Function]

When the APP successfully exits the connection interface, the APP sends this request so that the device knows that the APP is successfully exited.

[Request Interface on the APP]

http://192.168.169.1/app/exitrecorder

[Request Parameter on the APP]

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None.

[Return Parameter on the Device]

None.

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : "set success"
}
```

4.2.13 Enter the Exit Menu Mode

[Interface Function]

The APP is called when it enters the recorder setting interface or exits the recorder setting interface, so that the device can make corresponding processing.

[Request Interface on the APP]

http://192.168.169.1/app/setting

[Request Parameter on the APP]

Parameter	Type	Description
param	String	Corresponding value: ?param=enter or ?param=exit enter: Enter the menu mode exit: Exit the menu mode

[Return Parameter on the Device]

None.

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : "set success"
}
```

4.2.14 Enter the Exit Playback Mode

[Interface Function]

The APP is called when it enters the recorder setting interface or exits the recorder setting interface, so that the device can make corresponding processing.

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[Request Interface on the APP]

http://192.168.169.1/app/playback

[Request Parameter on the APP]

Parameter	Type	Description
param	String	Corresponding value: ?param=enter or ?param=exit enter: Enter the playback mode exit: Exit the playback mode

[Return Parameter on the Device]

None.

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : "set success"
}
```

4.3 Device File Management

4.3.1 File Identification

When the current device has multiple cameras, the APP can show which camera (front camera, internal camera, rear camera) each file (picture, video) is generated by. Here supports three identifications: “front”, “internal”, “rear”; the APP can identify the identification by some characters (case insensitive) of the file name in the file list.

- Front: The file name ends with “_f” or “_F”, for example: 2022080512444014_f.ts.
- Internal: The file name ends with “_i” or “_I”, for example: 2022080512444014_i.ts.
- Rear: The file name ends with “_b” or “_B”, for example: 2022080512444014_b.ts.

4.3.2 Obtain the File List (One-time Acquisition)

[Interface Function]

Obtain all information of pictures and videos on the device at one time, and list them by different storage paths.

[Request Interface on the APP]

http://192.168.169.1/app/getfilelist

[Request Parameter on the APP]

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None.

[Return Parameter on the Device]

Parameter	Type	Description
folder	String	Path name of folder. loop: Loop recording file; emr: Collision warning file; event: Photo file; park: Parking monitoring file.
count	int	Number of files in the folder.
files	Json Array	File information list in the folder.
name	String	Relative path of single file.
duration	int	File duration (in s). The duration of the picture is 0.
size	int	Size of file (in KB).
createtime	int	Creating time of file (in s). Unix timestamp.
createtimestr	String	Creating date of file (yyyyMMddHHmmss). For example: 20221202153657.
type	int	File type: 1-picture; 2-video.

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : [{
        "folder" : "loop",
        "count" : 2,
        "files" : [{
            "name" : "mnt/sdcard/VIDEO_F/20201120190300_f.ts",
            "duration" : 60,
            "size" : 120000000,
            "createtime" : 1593417777,
            "type" : 2
        },
        {
            "name" : "mnt/sdcard/VIDEO_F/20201120190300_f.ts",
            "duration" : 60,
            "size" : 120000000,
            "createtime" : 1593417777,
            "type" : 2
        }]
    },
    {
        "folder" : "emr",
        "count" : 56,
```

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```
        "files" : [{
            "name" : "mnt/sdcard/VIDEO_F/20201120190300_f.ts",
            "duration" : 120,
            "size" : 120000000,
            "createtime" : 1593416877,
            "type" : 2
        },
        ...
        {
            "name" : "mnt/sdcard/VIDEO_F/20201120190300_f.ts",
            "duration" : 120,
            "size" : 120000000,
            "createtime" : 1593416677,
            "type" : 2
        }]
    },
    {
        "folder" : "event",
        "count" : 10,
        "files" : [{
            "name" : "mnt/sdcard/PICTURE_F/20201120190300_f.jpg",
            "duration" : 0,
            "size" : 200,
            "createtime" : 1593416877,
            "type" : 1
        },
        ...
        {
            "name" : "mnt/sdcard/PICTURE_F/20201120190300_f.jpg",
            "duration" : 0,
            "size" : 200,
            "createtime" : 1593416677,
            "type" : 1
        }]
    },
    {
        "folder" : "park",
        "count" : 10,
        "files" : [{
            "name" : "mnt/sdcard/VIDEO_F/20201120190300_f_GAP.ts",
            "duration" : 120,
            "size" : 120000000,
            "createtime" : 1593416877,
            "type" : 2
        },
        ...
        {
            "name": "mnt/sdcard/VIDEO_F/20201120190300_f_GAP.ts",
            "duration" : 120,
            "size" : 120000000,
            "createtime" : 1593416677,
            "type" : 2
        }]
    }]
}
```

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CAUTION

You can return as many folders as you actually have, and you do not have to return any folders. For example, there is no park folder, the park field may not be included in the return.

4.3.3 Obtain the File List (Paging Acquisition)

[Interface Function]

Specify to obtain the folder and pagination information, and obtain the file list information of pictures and videos on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getfilelist

[Request Parameter on the APP]

Parameter	Type	Description
folder	String	Specified file type. For example: folder=loop. loop: Loop recording file; emr: Collision warning file; event: Photo file; park: Parking monitoring file.
start	int	Start index. For example: folder=loop&start=0, the index starts from 0.
end	int	End index. For example: folder=loop&start=0&end=99. the index starts from 0 and ends from 99.

[Return Parameter on the Device]

Parameter	Type	Description
folder	String	Path name of folder. loop: Loop recording file; emr: Collision warning file; event: Photo file; park: Parking monitoring file.
count	int	Number of files in the folder.
files	Json Array	File information list in the folder.

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Parameter	Type	Description
name	String	Relative path of single file.
duration	int	File duration (in s). The duration of the picture is 0.
size	int	Size of file (in KB).
createtime	int	Creating time of file (in s). Unix timestamp.
createtimestr	String	Creating date of file (yyyyMMddHHmmss). For example: 20221202153657.
type	int	File type: 1-picture; 2-video.

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : [{
        "folder" : "loop",
        "count" : 100,
        "files": [{
            "name" : "mnt/sdcard/VIDEO_F/20201120190300_f.ts",
            "duration": 60,
            "size" : 120000000,
            "createtime" : 1605870180,
            "createtimestr" : 20200629160257,
            "type" : 2
        },
        ...
        {
            "name" : "mnt/sdcard/VIDEO_F/20201120190300_b.ts",
            "duration" : 60,
            "size" : 64000000,
            "createtime" : 1605870180,
            "createtimestr" : 20200629160257,
            "type" : 2
        }]
    }]
}
```

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CAUTION

If the query parameter is incorrect, the folder does not exist, or the parameter exceeds the number of files, there is no file to return.

start: which index does it start from; end: which index does it end from.

For example 201 files, the size of each page is 100, the range of the first page is 0-99; the range of the second page is 100-199; the range of the third page is 200-200.

The data range obtained is [start, end], the start index starts from 0, and the paged data can be returned in reverse chronological order, the index of the latest file is the minimum.

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4.3.4 Obtain the File Thumbnail

[Interface Function]

Obtain the thumbnails of the pictures and videos on the device, so that the APP client can display the small pictures of the file before downloading the file.

[Request Interface on the APP]

http://192.168.169.1/app/getthumbnail

[Request Parameter on the APP]

Parameter	Type	Description
file	String	Relative path of a file, for example: ?file=/mnt/sdcard/VIDEO_F/20201120190300_f.ts

[Return Parameter on the Device]

None.

[Example of Device Return after Success]

The content of the corresponding file thumbnail is the file thumbnail downloaded by the device.

CAUTION

If the device fails to obtain the file thumbnail, the device can upload a fixed small picture instead. The moov in the file is stored in front of the mdat, so that the APP player can get the index format of the key frame in the file as soon as possible and display the first picture of the video as soon as possible.

4.3.5 Playback the File

4.3.5.1 Video Playback (Source File)

The current mainstream media files are packaged in mp4 format, and it is recommended that the device files should be in mp4 format (The media file format of the current device is ts, mp4 or mow), the playback mode used by APP can be ts, mp4 or mov while downloading and playing, and the requirements for the device are:

1. If the packaged format of a file is mp4, the moov in this file is stored in front of the mdat, so that the APP player can get the index format of the key frame in the file as soon as possible and display the first picture of the video as soon as possible.
2. The device needs to support the download protocol of HTTP file, and it is best to support Rang (The current device has supported Range).

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Source File of Video Playback	Note
Description	Transmission control of video playback source file (based on HTTP).
Interface	http://192.168.169.1:80/filepath，for example: http://192.168.169.1:80/mnt/sdcard/VIDEO_F/20230324143125_F.ts
Transmission layer	TCP Choose whether to use the source file for video playback according to the 6th bit of the device capability set.
HTTP version	HTTP/1.1
Method supported by HTTP	GET
Suffix of media package format supported	“.ts” or “.TS”; “.mov” or “.MOV”; “.mp4” or “.MP4”

4.3.5.2 Video Playback (Small Streaming)

Video Playback Small Streaming	Note
Description	Transmission control of video playback small streaming (based on RTSP).
Interface	rtsp://192.168.169.1:554/filepath or rtsp://192.168.169.1:8554/filepath, for example: rtsp://192.168.169.1:554/mnt/sdcard/VIDEO_F/20230324143125_F.ts.
Transmission layer	UDP or TCP（It is best to use TCP）. Choose whether to use the small streaming for video playback according to the 6th bit of the device capability set, or set the transport to **RTSP over UDP** or **RTSP over TCP**.
RTSP version	RTSP/1.0.
Method supported by RTSP	OPTIONS, DESCRIBE, SETUP, TEARDOWN, PLAY, PAUSE.
Preconditions for opening the RTSP server on the device	The APP sends a request to enter the playback mode: http://192.168.169.1/app/playback?param=enter
Media encoding formats supported by the pushflow on the device	Video: H.264/H.265; Audio: AAC.

4.3.6 Download the File

4.3.6.1 Video or Picture

[Interface Function]

Obtain the historical picture or video files on the device.

[Request Interface on the APP]

http://192.168.169.1/mnt/sdcard/VIDEO_F/20201120190300_f.ts

[Request Parameter on the APP]

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None.

[Return Parameter on the Device]

None.

[Example of Device Return after Success]

The content of the corresponding file is the historical picture or video files downloaded from the device.

4.3.6.2 GPS Data File

To display GPS data information at the same time, it is necessary to record the GPS data corresponding to the video file when playing the video online or downloading the video to the local mobile phone. Here, you can choose whether to capture the GPS data according to the 0 th bit of the device capability set.

4.3.6.2.1 Download the GPS File from GPSdata Folder

[Interface Function]

Download the GPS data file in the GPSdata folder by HTTP.

[Request Interface on the APP]

http://192.168.169.1/GPSdata/2023-07-06_16_25_59_f.TXT

[Request Parameter on the APP]

None.

[Return Parameter on the Device]

Parameters	Instances
Date occurred	2023/07/06 (Data format: yyyy/MM/dd)
Time occurred	16:25:59 (Time format: HH:mm:ss)
Latitude	N:22.524790 (N: Northern latitude, S: Southern latitude, unit: degree)
Longitude	E:113.935379 (E: East longitude，W: West longitude, unit: degree)
Speed	80.00
Unit of speed	km/h (Other unit needs to be converted to km/h)
Heading	271.80 (Unit: degree, parameter range: [0~360].)
Altitude	500.00 (Unit: m, parameter range: [-9999~9999].)
Number of satellites currently searched	17 (Unit: pcs. It is used to display the signal intensity of GPS.)
X-axis data of Gsensor	-0.004 (Unit: G)

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Parameters	Instances
Y-axis data of Gsensor	-0.137 (Unit: G)
Z-axis data of Gsensor	+0.054 (Unit: G)

[Example of Content Data]

- Valid data

```
2023/07/06 16:25:59 N:22.524790 E:113.935379 80.00 km/h 271.80 500.00 17 x:-0.004
    y:-0.137 z:+0.054
```

- Data with invalid fields (The current environment does not support, such as in the tunnel)

```
2023/07/06 16:25:59 NA NA NA km/h NA NA NA x:-0.004 y:-0.137 z:+0.054
```

- Not supported fields by the current GPS module

```
2023/07/06 16:25:59 N:22.524790 E:113.935379 80.00 km/h - - - x:-0.004 y:-0.137 z
    :+0.054
```

[Example of Device Return after Success]

```
2023/07/06 16:25:59 N:22.524790 E:113.935379 80.00 km/h 271.80 500.00 17 x:-0.004 y
    :-0.137 z:+0.054
2023/07/06 16:26:00 N:22.524790 E:113.935378 80.00 km/h 271.80 500.00 17 x:-0.004 y
    :-0.137 z:+0.054
2023/07/06 16:26:01 N:22.524790 E:113.935377 80.00 km/h 271.80 500.00 17 x:-0.004 y
    :-0.137 z:+0.054
2023/07/06 16:26:02 N:22.524790 E:113.935376 80.00 km/h 271.80 500.00 17 x:-0.004 y
    :-0.137 z:+0.054
2023/07/06 16:26:03 N:22.524790 E:113.935375 80.00 km/h 271.80 500.00 17 x:-0.004 y
    :-0.137 z:+0.054
2023/07/06 16:26:04 N:22.524790 E:113.935374 80.00 km/h 271.80 500.00 17 x:-0.004 y
    :-0.137 z:+0.054
2023/07/06 16:26:05 N:22.524790 E:113.935373 80.00 km/h 271.80 500.00 17 x:-0.004 y
    :-0.137 z:+0.054
2023/07/06 16:26:06 N:22.524790 E:113.935372 80.00 km/h 271.80 500.00 17 x:-0.004 y
    :-0.137 z:+0.054
2023/07/06 16:26:07 N:22.524790 E:113.935371 80.00 km/h 271.80 500.00 17 x:-0.004 y
    :-0.137 z:+0.054
```

4.3.6.2.2 Download the GPS File from Relative Path

[Interface Function]

The GPS data files are downloaded through the relative path, and when querying the file list, the relative path of the GPS file is returned through the “GPSPATH” node, such as:

```
{
    "name" : "mnt/sdcard/VIDEO_F/20201120190300_f.ts",
    "duration" : 60,
```

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```
    "size" : 120000000,
    "createtime" : 1593417777,
    "type" : 2,
    "GPSPATH" : "mnt/sdcard/VIDEO_F/20201120190300_f.TXT"
}
```

[Request Interface on the APP]

http://192.168.169.1/mnt/sdcard/VIDEO_F/20201120190300_f.TXT

[Request Parameter on the APP]

None.

[Return Parameter on the Device]

Parameters	Instances
Date occurred	2023/07/06 (Data format: yyyy/MM/dd)
Time occurred	16:25:59 (Time format: HH:mm:ss)
Latitude	N:22.524790 (N: North latitude, S: South latitude, unit: degree)
Longitude	E:113.935379 (E: East longitude, W: West longitude, unit: degree)
Speed	80.00
Unit of speed	km/h (Other unit needs to be converted to km/h)
Heading	271.80 (Unit: degree, parameter range: [0~360].)
Altitude	500.00 (Unit: m, parameter range: [-9999~9999].)
Number of satellites currently searched	17 (Unit: pcs. It is used to display the signal intensity of GPS.)
X-axis data of Gsensor	-0.004 (Unit: G)
Y-axis data of Gsensor	-0.137 (Unit: G)
Z-axis data of Gsensor	+0.054 (Unit: G)

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[Example of Content Data]

- Valid data

```
2023/07/06 16:25:59 N:22.524790 E:113.935379 80.00 km/h 271.80 500.00 17 x:-0.004
    y:-0.137 z:+0.054
```

- Data with invalid fields (The current environment does not support, such as in the tunnel)

```
2023/07/06 16:25:59 NA NA NA km/h NA NA NA x:-0.004 y:-0.137 z:+0.054
```

- Not supported fields by the current GPS module

```
2023/07/06 16:25:59 N:22.524790 E:113.935379 80.00 km/h - - - x:-0.004 y:-0.137 z
    :+0.054
```

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[Example of Device Return after Success]

```
2023/07/06 16:25:59 N:22.524790 E:113.935379 80.00 km/h 271.80 500.00 17 x:-0.004 y
    :-0.137 z:+0.054
2023/07/06 16:26:00 N:22.524790 E:113.935378 80.00 km/h 271.80 500.00 17 x:-0.004 y
    :-0.137 z:+0.054
2023/07/06 16:26:01 N:22.524790 E:113.935377 80.00 km/h 271.80 500.00 17 x:-0.004 y
    :-0.137 z:+0.054
2023/07/06 16:26:02 N:22.524790 E:113.935376 80.00 km/h 271.80 500.00 17 x:-0.004 y
    :-0.137 z:+0.054
2023/07/06 16:26:03 N:22.524790 E:113.935375 80.00 km/h 271.80 500.00 17 x:-0.004 y
    :-0.137 z:+0.054
2023/07/06 16:26:04 N:22.524790 E:113.935374 80.00 km/h 271.80 500.00 17 x:-0.004 y
    :-0.137 z:+0.054
2023/07/06 16:26:05 N:22.524790 E:113.935373 80.00 km/h 271.80 500.00 17 x:-0.004 y
    :-0.137 z:+0.054
2023/07/06 16:26:06 N:22.524790 E:113.935372 80.00 km/h 271.80 500.00 17 x:-0.004 y
    :-0.137 z:+0.054
2023/07/06 16:26:07 N:22.524790 E:113.935371 80.00 km/h 271.80 500.00 17 x:-0.004 y
    :-0.137 z:+0.054
```

4.3.6.2.3 Download the GPS Data Written in the Video File

[Interface Function]

Download the GPS data written in the video file (This is another way to save GPS data: the GPS data is written at the end of the video file, and there is a GPS data block every second).

[Request Interface on the APP]

http://192.168.169.1/mnt/sdcard/VIDEO_F/20201120190300_f.ts

[Request Parameter on the APP]

None.

[Data Structure]

The following data structure is recommended: Write the length of GPS data at the end of the video, which is convenient for mobile APP or computer application to quickly retrieve GPS data.

Header 4 bytes	Characteristic character of header 4 bytes	N bytes data	Characteristic character of tail 4 bytes	Tail 4 bytes
8+N+8	SKIP	(data header) + n of (data block)	&&&&	8+N+8

- Basic structure

  8 bytes (header) + N bytes + 8 bytes (tail)

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- Header information

  The first 4 bytes indicate the length, and the last 4 bytes indicate the segment names (ASCII, the name is fixed to "SKIP")

  4 bytes length

  Storage method of the first 4 bytes (length): Arrange according to high bits in the front and low bits in the back, for example: the length is 0x12345678, the storage method is **0x12 0x34 0x56 0x78**. 4 bytes length = length of data content + 8.

- Tail information

  The first 4 bytes are fixed to "&&&&", and the last 4 bytes are the length, which is the same with the first 4 bytes in the header information.

[Data Content Format]

Data Header	**Data Block 0**	**.....**	**Data Block n**
20 bytes	132 bytes	.....	132 character

[Data Header Format]

Data Flag	**Reserved Bit**	**Number of Data Blocks**
15 bytes: chIdentificaion	1 byte	4 bytes: nCount

- Data Header

  It contains the data identification, reserved bit, and number of data blocks.

- chIdentificaion

  It is fixed to a string:

```
LIGOGPSINFO
```

- nCount

  Total number of data blocks. One data block is generated per second, and the default value is 0. If 60 GPS data blocks are written at the end of the video, the nCount is 60. It is stored in hexadecimal mode, with the high bit in the back and the low bit in the front. For example:

```
0x3c 0x00 0x00 0x00
```

[Data Block Format]

The size of the data block is 132 bytes. The data block contains data block header and data block content.

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Data Block Header	Data Block Content
4 bytes: nIndex	128 bytes: chInformation

- nIndex

  It is 4 bytes. It is the index of the data block and increases from 1. The index of the last data block is equal to the number of data blocks (nCount) described in the data header. The storage method of nIndex is the same with that of the nCount.

- chInfomation

  It is fixed to 128 bytes. It is in the ASCII string form. It indicates the driver data, and the data format is as follows.

```
year/month/data hour: minute: second longitude latitude speed km/h X-axis-
    acceleration Y-axis-acceleration Z-axis-acceleration heading-angle altitude
    magnetic-declination
```

  The data need to be separated by a space, as shown below:

```
2023/07/06 16:25:59 N:22.524790 E:113.935379 80.0 km/h x:-0.04 y:-0.31 z:+0.54 A
    :271.8 H:500.0 M:E12.1
```

Parameters	Instances
Date occurred	2023/07/06 (Data format: yyyy/MM/dd)
Time occurred	16:25:59 (Time format: HH:mm:ss)
Latitude	N:22.524790 (N: North latitude, S: South latitude, unit: degree)
Longitude	E:113.935379 (E: East longitude, W: West longitude, unit: degree)
Speed	80.00
Unit of speed	km/h (Other unit needs to be converted to km/h)
X-axis data of Gsensor	-0.004 (Unit: G)
Y-axis data of Gsensor	-0.137 (Unit: G)
Z-axis data of Gsensor	+0.054 (Unit: G)
Heading	271.80 (Unit: degree, parameter range: [0~360].)
Altitude	500.00 (Unit: m, parameter range: [-9999~9999].)
Magnetic declination	E12.1

[Notice]

1. If the above data does not occupy 128 bytes, you need to fill it with a null character (0x00).

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2. To maintain compatibility, the GPS data must contain the fields listed in the above table. Fields that are not supported or are not available are filled according to the following rule.

   a. If a part of data is not supported, you can use the middle horizontal line ('-') instead, for example: the X/Y/Z-axis information of Gsensor are supported.

```
2023/07/06 16:25:59 N:22.524790 E:113.935379 80.0 km/h - - - A:271.8 H:500.0 M:
    E12.1
```

   b. If the current positioning is not successful, the longitude and latitude are all 0. For example: positioning failure or data invalid.

```
2023/07/06 16:25:59 N:0 E:0 0 km/h x:-0.04 y:-0.31 z:+0.54 A:0 H:0 M:0
```

3. The latitude (N: north latitude; S: south latitude) only keeps 6 decimal places. Degree format: such as, the obtained data by GPS module is 1122.34566.

```
Float value = 1122.34566/100(integer of degree obtained)+22.34566/60.0f(Convert
    minutes into degrees, and 60 minutes is equal to 1 Degree.)
```

4. The longitude (E: east longitude; W: west longitude) only keeps 6 decimal places. Degree format: such as, the obtained data by GPS module is 1122.34566.

```
Float value = 1122.34566/100(integer of degree obtained)+22.34566/60.0f(Convert
    minutes into degrees, and 60 minutes is equal to 1 Degree.)
```

5. The speed only keeps 1 decimal places. The unit is km/h. If the unit of the speed obtained by GPS module is n mile/h, the n mile/h needs to be converted to km/h (1 n mile/h=1.852 km/h). Note that the km/h is only an identification and does not indicate the unit of the speed.

6. The Gsensor only keeps 2 decimal places. The deflection angle in each direction is at most 2 decimal places. The unit is G. For example:

```
x:-0.04 y:-0.31 z:+0.54
```

7. Heading (refer to $GPRMC). It only keeps 1 decimal place. For example, A:271.8, A indicates heading data; the following data is angle (range: 0~360, the data is from $GPRMC and it does not need conversion).

8. Altitude (refer to $GPGGA). It only keeps 1 decimal place. "H" indicates altitude data; the following data is from $GPGGA and it does not need conversion.

9. Magnetic declination (refer to $GPRMC). It only keeps 1 decimal place. "M" indicates magnetic declination data; "E" indicates magnetic declination direction (E: east, W: west); the following data is from $GPRMC and it does not need conversion.

10. To obtain the GPS data written at the end of the video when playing back the video, the HTTP server on the device needs to support syntaxs such as "Range bytes=0-", which is convenient for the APP to download the GPS data on the device.

[Example of Device Return Parameter after Success]

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Corresponding GPS data content.

4.3.7 Delete the File

[Interface Function]

Delete the file on the device. The file is queried by using "obtain the file list" interface.

[Request Interface on the APP]

http://192.168.169.1/app/deletefile

[Request Parameter on the APP]

Parameter	Type	Description
file	String	Relative path of the file, for example: ?file=/mnt/sdcard/VIDEO_F/20201120190300_f.ts

[Return Parameter on the Device]

None.

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : "delete success"
}
```

4.3.8 Upload the File

[Interface Function]

It is mainly used for uploading files such as bin or pkg required by OTA on the background.

[Request Type on the APP]

POST.

[Request Interface on the APP]

http://192.168.169.1/upload

[Request Parameter on the APP]

File path saved to device/actual byte size of file. For example: /mnt/sdcard/8388622.

[Example of Device Return after Success]

```
HTTP/1.1 200 OK<CRLF>
Accept-Encoding: identity<CRLF>
Connection: Keep-Alive<CRLF>
Black Line
Main body
```

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[Example of Device Return after Failure]

```
1. TF card or SD card has been pulled out:
    {
        "result" : 2,
        "info" : "upload fail"
    }

2. TF card or SD card is not formatted:
    :
        "result" : 3,
        "info" : "upload fail"
    }
```

4.4 Normal Setting Item

4.4.1 Query the Parameter List of Each Setting Item on the Device

[Interface Function]

Query the parameter list of each setting item on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamitems

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=all: Query the parameter list of each setting item.

[Return Parameter on the Device]

Parameter	Type	Description
name	String	Name of the setting item. For example: mic, it is used to configure the switch of the microphone.
items	Json Array(String)	A readable parameter list corresponding to this setting item. For example, the mic has two settable parameters: on: open off: close
index	Json Array(int)	List of parameter values corresponding to this setting item, which is used to set the parameter values of the mic. For example, the mic has two settable parameters: 1: on

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Parameter	Type	Description
		0: off

[Example of Device Return after Success]


```
{
    "result" : 0,
    "info" : [{
        "name" : "mic", /* MIC recorder */
        "items" : ["off", "on"],
        "index" : [0, 1]
    },
    {
        "name" : "osd", /* Time watermark */
        "items" : ["off", "on"],
        "index" : [0, 1]
    },
    {
        "name" : "logo_osd", /* Logo watermark */
        "items" : ["off", "on"],
        "index" : [0, 1]
    },
    {
        "name" : "rec_resolution", /* Recoding resolution */
        "items" : ["720P", "1080P", "1296P", "2K"],
        "index" : [0, 1, 2, 3]
    },
    {
        "name" : "rec_split_duration", /* Loop video recording (duration of recording
     file) */
        "items" : ["1MIN", "2MIN"],
        "index" : [0, 1]
    },
    {
        "name" : "encodec", /* Video encoding format */
        "items" : ["h.264", "h.265"],
        "index" : [0, 1]
    },
    {
        "name" : "speaker", /* Speaker volume */
        "items" : ["off", "low", "middle", "high", "very high"],
        "index" : [0, 1, 2, 3, 4]
    },
    {
        "name" : "gsr_sensitivity", /* Gsensor (collision) sensitivity */
        "items" : ["high", "low", "off"],
        "index" : [0, 1, 2]
    },
    {
        "name" : "rec", /* Recording status */
        "items" : ["off", "on"],
        "index" : [0, 1]
    },
    {
```


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```
        "name" : "language", /* Language */
        "items" : ["en_US", "zh_CN", "zh_CHT", "ja", "ru", "th", "cran"],
        "index" : [0, 1, 2, 3, 4, 5, 6]
    },
    {
        "name" : "timelapse_rate", /* Frame rate of time-lapse video recording */
        "items" : ["off", "1fps", "2fps", "5fps"],
        "index" : [0,1,2,3]
    },
    {
        "name" : "park_record_time", /* Duration of parking monitoring recording */
        "items" : ["off", "12hour", "24hour", "48hour"],
        "index" : [0, 1, 2, 3]
    },
    {
        "name" : "park_gsr_sensitivity", /* Parking monitoring collision sensitivity
*/
        "items" : ["high","low","off"],
        "index" : [0, 1, 2]
    },
    {
        "name" : "parking_mode", /* Parking monitoring mode */
        "items" : ["off", "timelapse", "normrec"],
        "index" : [0, 1, 2]
    },
    {
        "name" : "parking_monitor", /* Parking monitoring switch */
        "items" : ["off", "on"],
        "index" : [0, 1]
    },
    {
        "name" : "light_fre", /* Light frequency */
        "items" : ["50Hz", "60Hz"],
        "index" : [0, 1]
    },
    {
        "name" : "rear_mirror", /* Rear mirror switch */
        "items" : ["off", "on"],
        "index" : [0, 1]
    },
    {
        "name" : "wdr", /* Wide dynamic switch */
        "items" : ["off", "on"],
        "index" : [0, 1]
    },
    {
        "name" : "voice_control", /* Voice control switch */
        "items" : ["off", "on"],
        "index" : [0, 1]
    },
    {
        "name" : "screen_standby", /* Screen standby duration */
        "items" : ["off", "10s", "30s", "60s"],
        "index" : [0, 1, 2, 3]
    },
    {
```

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```
        "name" : "auto_poweroff", /* Auto poweroff duration */
        "items" : ["off", "1H", "12H"],
        "index" : [0, 1, 2]
    },
    {
        "name" : "adas", /* ADAS switch */
        "items" : ["off", "on"],
        "index" : [0, 1]
    },
    {
        "name" : "boot_sound", /* Boot sound switch */
        "items" : ["off", "on"],
        "index" : [0, 1]
    },
    {
        "name" : "video_filp", /* Video vertical flip switch */
        "items" : ["off", "on"],
        "index" : [0, 1]
    },
    {
        "name" : "video_mirror", /* Video horizontal mirror switch */
        "items" : ["off", "on"],
        "index" : [0, 1]
    },
    {
        "name" : "key_tone", /* Key tone switch */
        "items" : ["off", "on"],
        "index" : [0, 1]
    }]
}
```

4.4.2 Parameter List of Microphone

[Interface Function]

Query the setting item and parameter list of the microphone (MIC) on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamitems

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=mic: Obtain the parameter list of the MIC switch.

[Return Parameter on the Device]

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Parameter	Type	Description
items	Json Array(String)	A readable parameter list corresponding to this setting item. For example, the mic has two settable parameters: on: open off: close
index	Json Array(int)	List of parameter values corresponding to this setting item, which is used to set the parameter values of the mic. For example, the mic has two settable parameters: 1: on 0: off

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "items" : ["off", "on"],
        "index" : [0, 1]
    }
}
```

4.4.3 Parameter List of Time Watermark

[Interface Function]

Query the setting item and parameter list of the time watermark on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamitems

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=osd: Obtain the parameter list of the time watermark switch.

[Return Parameter on the Device]

Parameter	Type	Description
items	Json Array(String)	A readable parameter list corresponding to this setting item. For example, the osd has two settable parameters:

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Parameter	Type	Description
		on: open off: close
index	Json Array(int)	List of parameter values corresponding to this setting item, which is used to set the parameter values of the osd. For example, the osd has two settable parameters: 1: on 0: off

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "items" : ["off", "on"],
        "index" : [0, 1]
    }
}
```

4.4.4 Parameter List of Logo Watermark

[Interface Function]

Query the setting item and parameter list of the logo watermark on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamitems

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=logo_osd: Obtain the parameter list of the logo watermark switch.

[Return Parameter on the Device]

Parameter	Type	Description
items	Json Array(String)	A readable parameter list corresponding to this setting item. For example, the logo_osd has two settable parameters: on: open off: close

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Parameter	Type	Description
index	Json Array(int)	List of parameter values corresponding to this setting item, which is used to set the parameter values of the logo_osd. For example, the logo_osd has two settable parameters: 1: on 0: off

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "items":["off", "on"],
        "index":[0, 1]
    }
}
```

4.4.5 Parameter List of Recording Resolution

[Interface Function]

Query the setting item and parameter list of the recording resolution on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamitems

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=rec_resolution: Obtain the parameter list of the recording resolution.

[Return Parameter on the Device]

Parameter	Type	Description
items	Json Array(String)	A readable parameter list corresponding to this setting item. For example, the rec_resolution has four settable parameters: “720P”, “1080P”, “1296P”, “2K”.
index	Json Array(int)	List of parameter values corresponding to this setting item, which is used to set the parameter values of the rec_resolution. For example, the rec_resolution has four settable parameters: 0: 720P

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Parameter	Type	Description
		1: 1080P 2: 1296P 3: 2K

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "items":["720P", "1080P", "1296P", "2K"],
        "index":[0, 1, 2, 3]
    }
}
```

4.4.6 Parameter List of Recording File Duration

[Interface Function]

Query the setting item and parameter list of the recording file duration on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamitems

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=rec_split_duration: Obtain the parameter list of the recording file duration.

[Return Parameter on the Device]

Parameter	Type	Description
items	Json Array(String)	A readable parameter list corresponding to this setting item. For example, the rec_split_duration has two settable parameters: “1MIN”, “2MIN”.
index	Json Array(int)	List of parameter values corresponding to this setting item, which is used to set the parameter values of the rec_split_duration. For example, the rec_split_duration has two settable parameters: 0: 1MIN 1: 2MIN

[Example of Device Return after Success]

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```
{
    "result" : 0,
    "info" : {
        "items" : ["1MIN", "2MIN"],
        "index" : [0, 1]
    }
}
```

4.4.7 Parameter List of Video Encoding Format

[Interface Function]

Query the setting item and parameter list of the video encoding format on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamitems

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=encodec: Obtain the parameter list of the video encoding format.

[Return Parameter on the Device]

Parameter	Type	Description
items	Json Array(String)	A readable parameter list corresponding to this setting item. For example, the encodec has two settable parameters: “h.264”, “h.265”.
index	Json Array(int)	List of parameter values corresponding to this setting item, which is used to set the parameter values of the encodec. For example, the encodec has two settable parameters: 0: h.264 1: h.265

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "items" : ["h.264", "h.265"],
        "index" : [0, 1]
    }
}
```

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4.4.8 Parameter List of Speaker Volume

[Interface Function]

Query the setting item and parameter list of the speaker volume on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamitems

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=speaker: Query the parameter list of the speaker volume.

[Return Parameter on the Device]

Parameter	Type	Description
items	Json Array(String)	A readable parameter list corresponding to this setting item. For example, the speaker has five settable parameters: “off”, “low”, “middle”, “high”, “very high”.
index	Json Array(int)	List of parameter values corresponding to this setting item, which is used to set the parameter values of the speaker. For example, the speaker has five settable parameters: 0, 1, 2, 3, 4.

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "items" : ["off", "low", "middle", "high", "very high"],
        "index" : [0, 1, 2, 3, 4]
    }
}
```

4.4.9 Parameter List of Collision Sensitivity

[Interface Function]

Query the setting item and parameter list of the collision sensitivity on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamitems

[Request Parameter on the APP]

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Parameter	Type	Description
param	String	?param=gsr_sensitivity: Obtain the parameter list of the collision sensitivity.

[Return Parameter on the Device]

Parameter	Type	Description
items	Json Array(String)	A readable parameter list corresponding to this setting item. For example, the gsr_sensitivity has three settable parameters: “high”, “low”, “off”.
index	Json Array(int)	List of parameter values corresponding to this setting item, which is used to set the parameter values of the gsr_sensitivity. For example, the gsr_sensitivity has three settable parameters: 0: high 1: low 2: off

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "items" : ["high", "low", "off"],
        "index" : [0, 1, 2]
    }
}
```

4.4.10 Parameter List of Recording Status

[Interface Function]

Query the setting item and parameter list of the recording status on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamitems

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=rec: Query the parameter list of the recording status.

[Return Parameter on the Device]

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Parameter	Type	Description
items	Json Array(String)	A readable parameter list corresponding to this setting item. For example, the rec has two settable parameters: “off”, “on”.
index	Json Array(int)	List of parameter values corresponding to this setting item, which is used to set the parameter values of the rec. For example, the rec has two settable parameters: 0: off 1: on

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "items" : ["off", "on"],
        "index" : [0, 1]
    }
}
```

4.4.11 Parameter List of Language Type

[Interface Function]

Query the setting item and parameter list of the language type on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamitems

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=language: Obtain the parameter list of the language type.

[Return Parameter on the Device]

Parameter	Type	Description
items	Json Array(String)	A readable parameter list corresponding to this setting item. For example, the language has seven settable parameters: “en_US”, “zh_CN”, “zh_CHT”, “ja”, “ru”, “th”, “cran”.

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Parameter	Type	Description
index	Json Array(int)	List of parameter values corresponding to this setting item, which is used to set the parameter values of the language. For example, the language has seven settable parameters: 0: en_US 1: zh_CN (Simplified Chinese) 2: zh_CHT (Traditional Chinese) 3: ja (Japanese) 4: ru (Russian) 5: th (Thai) 6: cran (Vietnamese)

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "items" : ["en_US", "zh_CN", "zh_CHT", "ja", "ru", "th", "cran"],
        "index" : [0, 1, 2, 3, 4, 5, 6]
    }
}
```

4.4.12 Parameter List of Timelapse Recording Frame Rate

[Interface Function]

Query the setting item and parameter list of the timelapse recording frame rate on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamitems

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=timelapse_rate: Obtain the parameter list of the timelapse recording frame rate.

[Return Parameter on the Device]

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Parameter	Type	Description
items	Json Array(String)	A readable parameter list corresponding to this setting item. For example, the timelapse_rate has four settable parameters: “off”, “1fps”, “2fps”, “5fps”
index	Json Array(int)	List of parameter values corresponding to this setting item, which is used to set the parameter values of the timelapse_rate. For example, the timelapse_rate has four settable parameters: 0: off 1: 1fps 2: 2fps 3: 5fps

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "items" : ["off", "1fps", "2fps", "5fps"],
        "index" : [0, 1, 2, 3]
    }
}
```

4.4.13 Parameter List of Parking Monitoring Recording Duration

[Interface Function]

Query the setting item and parameter list of the parking monitoring recording duration on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamitems

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=park_record_time: Obtain the parameter list of the parking monitoring recording duration.

[Return Parameter on the Device]

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Parameter	Type	Description
items	Json Array(String)	A readable parameter list corresponding to this setting item. For example, the park_record_time has four settable parameters: off: Disabled 12hour: 12 hours 24hour: 24 hours 48hour: 48 hours
index	Json Array(int)	List of parameter values corresponding to this setting item, which is used to set the parameter values of the park_record_time. For example, the park_record_time has four settable parameters: 0: off 1: 12hour 2: 24hour 3: 48hour

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "items" : ["off", "12hour", "24hour", "48hour"],
        "index" : [0, 1, 2, 3]
    }
}
```

4.4.14 Parameter List of Parking Monitoring Collision Sense

[Interface Function]

Query the setting item and parameter list of the parking monitoring collision sense on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamitems

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=park_gsr_sensitivity: Obtain the parameter list of the parking monitoring collision sense.

[Return Parameter on the Device]

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Parameter	Type	Description
items	Json Array(String)	A readable parameter list corresponding to this setting item. For example, the park_gsr_sensitivity has three settable parameters: “high”, “low”, “off”.
index	Json Array(int)	List of parameter values corresponding to this setting item, which is used to set the parameter values of the park_gsr_sensitivity. For example, the park_gsr_sensitivity has three settable parameters: 0: high 1: low 2: off

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "items" : ["high", "low", "off"],
        "index" : [0, 1, 2]
    }
}
```

4.4.15 Parameter List of Parking Monitoring Mode

[Interface Function]

Query the setting item and parameter list of the parking monitoring mode on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamitems

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=parking_mode: Obtain the parameter list of the parking monitoring mode.

[Return Parameter on the Device]

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Parameter	Type	Description
items	Json Array(String)	A readable parameter list corresponding to this setting item. For example, the parking_mode has three settable parameters: “off”, “timelapse”, “normrec”.
index	Json Array(int)	List of parameter values corresponding to this setting item, which is used to set the parameter values of the parking_mode. For example, the parking_mode has three settable parameters: 0: off 1: timelapse 2: normrec (normal recording)

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "items" : ["off", "timelapse", "normrec"],
        "index" : [0, 1, 2]
    }
}
```

4.4.16 Parameter List of Parking Monitoring Switch

[Interface Function]

Query the setting item and parameter list of the parking monitoring switch on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamitems

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=parking_monitor: Obtain the parameter list of the parking monitoring switch.

[Return Parameter on the Device]

Parameter	Type	Description
items	Json Array(String)	A readable parameter list corresponding to this setting item. For example, the parking_monitor has two settable parameters: on, off.

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Parameter	Type	Description
index	Json Array(int)	List of parameter values corresponding to this setting item, which is used to set the parameter values of the parking_monitor. For example, the parking_monitor has two settable parameters: 1: on 0: off

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "items" : ["off", "on"],
        "index" : [0, 1]
    }
}
```

4.4.17 Parameter List of Light Frequency

[Interface Function]

Query the setting item and parameter list of the light frequency on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamitems

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=light_fre: Obtain the parameter list of the light frequency.

[Return Parameter on the Device]

Parameter	Type	Description
items	Json Array(String)	A readable parameter list corresponding to this setting item. For example, the light_fre has two settable parameters: 50Hz (Standard frequency of most countries in Europe and Asia), 60Hz (Standard frequency in North America and some South American countries)
index	Json Array(int)	List of parameter values corresponding to this setting item, which is used to set the parameter values of the light_fre. For example, the light_fre has two settable parameters:

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Parameter	Type	Description
		0: 50Hz 1: 60Hz

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "items" : ["50Hz", "60Hz"],
        "index" : [0, 1]
    }
}
```

4.4.18 Parameter List of Rear Mirror Switch

[Interface Function]

Query the setting item and parameter list of the rear mirror switch on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamitems

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=rear_mirror: Obtain the parameter list of the rear mirror switch.

[Return Parameter on the Device]

Parameter	Type	Description
items	Json Array(String)	A readable parameter list corresponding to this setting item. For example, the rear_mirror has two settable parameters: on, off.
index	Json Array(int)	List of parameter values corresponding to this setting item, which is used to set the parameter values of the rear_mirror. For example, the rear_mirror has two settable parameters: 1: on 0: off

[Example of Device Return after Success]

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```
{
    "result" : 0,
    "info" : {
        "items" : ["off", "on"],
        "index" : [0, 1]
    }
}
```

4.4.19 Parameter List of Wide Dynamic Switch

[Interface Function]

Query the setting item and parameter list of the wide dynamic switch on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamitems

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=wdr: Obtain the parameter list of the wide dynamic switch.

[Return Parameter on the Device]

Parameter	Type	Description
items	Json Array(String)	A readable parameter list corresponding to this setting item. For example, the wdr has two settable parameters: on, off.
index	Json Array(int)	List of parameter values corresponding to this setting item, which is used to set the parameter values of the wdr. For example, the wdr has two settable parameters: 1: on 0: off

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "items" : ["off", "on"],
        "index" : [0, 1]
    }
}
```

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4.4.20 Parameter List of Voice Control Switch

[Interface Function]

Query the setting item and parameter list of the voice control switch on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamitems

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=voice_control: Obtain the parameter list of the voice control switch.

[Return Parameter on the Device]

Parameter	Type	Description
items	Json Array(String)	A readable parameter list corresponding to this setting item. For example, the voice_control has two settable parameters: on, off.
index	Json Array(int)	List of parameter values corresponding to this setting item, which is used to set the parameter values of the voice_control. For example, the voice_control has two settable parameters: 1: on 0: off

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "items" : ["off", "on"],
        "index" : [0, 1]
    }
}
```

4.4.21 Parameter List of Screen Standby Duration

[Interface Function]

Query the setting item and parameter list of the screen standby duration on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamitems

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[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=screen_standby: Obtain the parameter list of the screen standby duration.

[Return Parameter on the Device]

Parameter	Type	Description
items	Json Array(String)	A readable parameter list corresponding to this setting item. For example, the screen_standby has four settable parameters: off, 10s, 30s, 60s.
index	Json Array(int)	List of parameter values corresponding to this setting item, which is used to set the parameter values of the screen_standby. For example, the screen_standby has four settable parameters: 0: off 1: 10s 2: 30s 3: 60s

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "items" : ["off", "10s", "30s", "60s"],
        "index" : [0, 1, 2, 3]
    }
}
```

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4.4.22 Parameter List of Auto Poweroff Duration

[Interface Function]

Query the setting item and parameter list of the auto poweroff duration on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamitems

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=auto_poweroff: Obtain the parameter list of the auto poweroff duration.

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[Return Parameter on the Device]

Parameter	Type	Description
items	Json Array(String)	A readable parameter list corresponding to this setting item. For example, the auto_poweroff has three settable parameters: off: Disabled 1H: 1 hour 12H: 12 hours
index	Json Array(int)	List of parameter values corresponding to this setting item, which is used to set the parameter values of the auto_poweroff. For example, the auto_poweroff has three settable parameters: 0: off 1: 1H 2: 12H

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "items" : ["off", "1H", "12H"],
        "index" : [0, 1, 2]
    }
}
```

4.4.23 Parameter List of ADAS Switch

[Interface Function]

Query the setting item and parameter list of the ADAS switch on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamitems

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=adas: Obtain the parameter list of the ADAS switch.

[Return Parameter on the Device]

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Parameter	Type	Description
items	Json Array(String)	A readable parameter list corresponding to this setting item. For example, the adas has two settable parameters: on, off.
index	Json Array(int)	List of parameter values corresponding to this setting item, which is used to set the parameter values of the adas. For example, the adas has two settable parameters: 1: on 0: off

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "items" : ["off", "on"],
        "index" : [0, 1]
    }
}
```

4.4.24 Parameter List of Boot Sound Switch

[Interface Function]

Query the setting item and parameter list of the boot sound switch on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamitems

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=boot_sound: Obtain the parameter list of the boot sound switch.

[Return Parameter on the Device]

Parameter	Type	Description
items	Json Array(String)	A readable parameter list corresponding to this setting item. For example, the boot_sound has two settable parameters: on, off.
index	Json Array(int)	List of parameter values corresponding to this setting item, which is used to set the parameter values of the boot_sound. For example, the boot_sound has two settable parameters:

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Parameter	Type	Description
		1: on 0: off

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "items" : ["off", "on"],
        "index" : [0, 1]
    }
}
```

4.4.25 Parameter List of Video Vertical Flip Switch

[Interface Function]

Query the setting item and parameter list of the video vertical flip switch on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamitems

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=video_filp: Obtain the parameter list of the video vertical flip switch.

[Return Parameter on the Device]

Parameter	Type	Description
items	Json Array(String)	A readable parameter list corresponding to this setting item. For example, the video_filp has two settable parameters: on, off.
index	Json Array(int)	List of parameter values corresponding to this setting item, which is used to set the parameter values of the video_filp. For example, the video_filp has two settable parameters: 1: on 0: off

[Example of Device Return after Success]

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```
{
    "result" : 0,
    "info" : {
        "items" : ["off", "on"],
        "index" : [0, 1]
    }
}
```

4.4.26 Parameter List of Video Horizontal Mirror Switch

[Interface Function]

Query the setting item and parameter list of the video horizontal mirror switch on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamitems

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=video_mirror: Obtain the parameter list of the video horizontal mirror switch.

[Return Parameter on the Device]

Parameter	Type	Description
items	Json Array(String)	A readable parameter list corresponding to this setting item. For example, the video_mirror has two settable parameters: on, off.
index	Json Array(int)	List of parameter values corresponding to this setting item, which is used to set the parameter values of the video_mirror. For example, the video_mirror has two settable parameters: 1: on 0: off

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "items" : ["off", "on"],
        "index" : [0, 1]
    }
}
```

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4.4.27 Parameter List of Key Tone Switch

[Interface Function]

Query the setting item and parameter list of the key tone switch on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamitems

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=key_tone: Obtain the parameter list of the key tone switch.

[Return Parameter on the Device]

Parameter	Type	Description
items	Json Array(String)	A readable parameter list corresponding to this setting item. For example, the key_tone has two settable parameters: on, off.
index	Json Array(int)	List of parameter values corresponding to this setting item, which is used to set the parameter values of the key_tone. For example, the key_tone has two settable parameters: 1: on 0: off

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "items" : ["off", "on"],
        "index" : [0, 1]
    }
}
```

4.4.28 All Current Values

[Interface Function]

Each setting item on the device side has a current value. If the user needs to know the current value, this interface is used to query the current value of all setting items.

[Request Interface on the APP]

http://192.168.169.1/app/getparamitems

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[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=all: Obtain the current value of all setting items.

[Return Parameter on the Device]

Parameter	Type	Description
value	int	For example, the parameter value corresponding to the setting item speaker corresponds to the value in the index array in the parameter list: 0: off 1: low 2: middle 3: high 4: very high

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : [{
        "name" : "mic", /* MIC recording */
        "value" : 0
    },
    {
        "name" : "osd", /* Time watermark */
        "value" : 1
    },
    {
        "name" : " logo_osd", /* Logo watermark */
        "value" : 1
    },
    {
        "name" : "rec_split_duration", /* Loop video (recording file duration) */
        "value" : 0
    },
    {
        "name" : "speaker", /* Speaker volume */
        "value" : 2
    },
    {
        "name" : "gsr_sensitivity", /* Gsensor (collision) sensitivity */
        "value" : 2
    },
    {
        "name" : "rec_resolution", /* Recording resolution */
        "value" : 1
```

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```
    },
    {
        "name" : "encodec", /* Video encoding format */
        "value" : 0
    },
    {
        "name" : "rec", /* Recording status */
        "value" : 0
    },
    {
        "name" : "language", /* Language */
        "value" : 0
    },
    {
        "name" : "timelapse_rate", /* Timelapse video frame rate */
        "value" : 0
    }
    ,
    {
        "name" : "park_record_time", /* Parking monitoring recording duration */
        "value" : 0
    },
    {
        "name" : "park_gsr_sensitivity", /* Parking monitoring collision sensitivity
*/
        "value" : 0
    },
    {
        "name" : "parking_mode", /* Parking monitoring mode */
        "value" : 0
    },
    {
        "name" : "parking_monitor", /* Parking monitoring switch */
        "value" : 0
    },
    {
        "name" : "light_fre", /* Light frequency */
        "value" : 0
    },
    {
        "name" : "rear_mirror", /* Rear mirror switch */
        "value" : 0
    }
    {
        "name" : "wdr", /* Wide dynamic switch */
        "value" : 0
    },
    {
        "name" : "voice_control", /* Voice control switch */
        "value" : 0
    },
    {
        "name" : "screen_standby", /* Screen standby duration */
        "value" : 0
    },
    {
```

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```
        "name" : "auto_poweroff", /* Auto poweroff duration */
        "value" : 0
    },
    {
        "name" : "adas", /* ADAS switch */
        "value" : 0
    },
    {
        "name" : "boot_sound", /* Boot sound switch */
        "value" : 0
    },
    {
        "name" : "video_filp", /* Video vertical flip switch */
        "value" : 0
    },
    {
        "name" : "video_mirror", /* Video horizontal mirror switch */
        "value" : 0
    },
    {
        "name" : "key_tone", /* Key tone switch */
        "value" : 0
    }]
}
```

NOTE

If you want to query a setting item instead of all settings, change the parameter param=all to the corresponding setting. For example, query the current value of the speaker: http://192.168.169.1/app/getparamvalue?param=speaker

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "value" : 3
    }
}
```

4.4.29 Current Value of Microphone Switch

[Interface Function]

Query the current value of the setting item of the microphone switch on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamvalue

[Request Parameter on the APP]

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Parameter	Type	Description
param	String	?param=mic: Obtain the current value of the microphone switch.

[Return Parameter on the Device]

Parameter	Type	Description
value	int	The parameter value corresponding to the setting item mic corresponds to the value in the index array in the parameter list: 0: off 1: on

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "value" : 1
    }
}
```

4.4.30 Current Value of Time Watermark Switch

[Interface Function]

Query the current value of the setting item of the time watermark switch on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamvalue

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=osd: Obtain the current value of the time watermark switch.

[Return Parameter on the Device]

Parameter	Type	Description
value	int	The parameter value corresponding to the setting item osd corresponds to the value in the index array in the parameter list: 0: off

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Parameter	Type	Description
		1: on

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "value" : 1
    }
}
```

4.4.31 Current Value of Logo Watermark Switch

[Interface Function]

Query the current value of the setting item of the logo watermark switch on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamvalue

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=logo_osd: Obtain the current value of the the logo watermark switch.

[Return Parameter on the Device]

Parameter	Type	Description
value	int	The parameter value corresponding to the setting item logo_osd corresponds to the value in the index array in the parameter list: 0: off 1: on

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info": {
        "value" : 1
    }
}
```

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4.4.32 Current Value of Recording Resolution

[Interface Function]

Query the current value of the setting value of the recording resolution on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamvalue

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=rec_resolution: Obtain the current value of the recording resolution.

[Return Parameter on the Device]

Parameter	Type	Description
value	int	The parameter value corresponding to the setting item rec_resolution corresponds to the value in the index array in the parameter list: 0: 720P 1: 1080P 2: 1296P 3: 2K

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "value" : 1
    }
}
```

4.4.33 Current Value of Recording File Duration

[Interface Function]

Query the current value of the setting item of the recording file duration on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamvalue

[Request Parameter on the APP]

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Parameter	Type	Description
param	String	?param=rec_split_duration: Obtain the current value of the recording file duration.

[Return Parameter on the Device]

Parameter	Type	Description
value	int	The parameter value corresponding to the setting item rec_split_duration corresponds to the value in the index array in the parameter list: 0: 1 minute 1: 2 minutes

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "value" : 1
    }
}
```

4.4.34 Current Value of Video Encoding Format

[Interface Function]

Query the current value of the setting item of the video encoding format on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamvalue

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=encodec: Obtain the current value of the video encoding format.

[Return Parameter on the Device]

Parameter	Type	Description
value	int	The parameter value corresponding to the setting item encodec corresponds to the value in the index array in the parameter list: 0: h.264

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Parameter	Type	Description
		1: h.265

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "value" : 1
    }
}
```

4.4.35 Current Value of Speaker Volume

[Interface Function]

Query the current value of the setting item of the speaker volume on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamvalue

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=speaker: Obtain the current value of the speaker volume.

[Return Parameter on the Device]

Parameter	Type	Description
value	int	The parameter value corresponding to the setting item speaker corresponds to the value in the index array in the parameter list: 0: off 1: low 2: middle 3: high 4: very high

[Example of Device Return after Success]

```
{
    "result" : 0,
```

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```
    "info" : {
        "value" : 1
    }
}
```

4.4.36 Current Value of Collision Sensitivity

[Interface Function]

Query the current value of the setting item of the collision sensitivity on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamvalue

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=gsr_sensitivity: Obtain the current value of the collision sensitivity.

[Return Parameter on the Device]

Parameter	Type	Description
value	int	The parameter value corresponding to the setting item gsr_sensitivity corresponds to the value in the index array in the parameter list: 0: high 1: low 2: off

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
    "value" : 1
    }
}
```

4.4.37 Current Value of Recording Status

[Interface Function]

Query the current value of the setting item of the recording status on the device.

[Request Interface on the APP]

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http://192.168.169.1/app/getparamvalue

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=rec: Obtain the current value of the recording status.

[Return Parameter on the Device]

Parameter	Type	Description
value	int	The parameter value corresponding to the setting item rec corresponds to the value in the index array in the parameter list: 1: on 0: off

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "value" : 1
    }
}
```

4.4.38 Current Value of Language Type

[Interface Function]

Query the current value of the setting item of the language type on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamvalue

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=language: Obtain the current value of the language type.

[Return Parameter on the Device]

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Parameter	Type	Description
value	int	The parameter value corresponding to the setting item language corresponds to the value in the index array in the parameter list: 0: en_US 1: zh_CN (Simplified Chinese) 2: zh_CHT (Traditional Chinese) 3: ja (Japanese) 4: ru (Russian) 5: th (Thai) 6: cran (Vietnamese)

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "value" : 1
    }
}
```

4.4.39 Current Value of Timelapse Recording Frame Rate

[Interface Function]

Query the current value of the setting item of the timelapse recording frame rate on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamvalue

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=timelapse_rate: Obtain the current value of the timelapse recording frame rate.

[Return Parameter on the Device]

Parameter	Type	Description
value	int	The parameter value corresponding to the setting item timelapse_rate corresponds to the value in the index array in the parameter list:

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Parameter	Type	Description
		0: off 1: 1fps 2: 2fps 3: 5fps

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "value" : 1
    }
}
```

4.4.40 Current Value of Parking Monitoring Recording Duration

[Interface Function]

Query the current value of the setting item of the parking monitoring recording duration on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamvalue

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=park_record_time: Obtain the current value of the parking monitoring recording duration.

[Return Parameter on the Device]

Parameter	Type	Description
value	int	The parameter value corresponding to the setting item park_record_time corresponds to the value in the index array in the parameter list: 0: off 1: 12hour 2: 24hour 3: 48hour

[Example of Device Return after Success]

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```
{
    "result" : 0,
    "info" : {
        "value" : 1
    }
}
```

4.4.41 Current Value of Parking Monitoring Collision Sensitivity

[Interface Function]

Query the current value of the setting item of the parking monitoring collision sensitivity.

[Request Interface on the APP]

http://192.168.169.1/app/getparamvalue

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=park_gsr_sensitivity: Obtain the current value of the parking monitoring collision sensitivity.

[Return Parameter on the Device]

Parameter	Type	Description
value	int	The parameter value corresponding to the setting item park_gsr_sensitivity corresponds to the value in the index array in the parameter list: 0: high 1: low 2: off

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "value" : 1
    }
}
```

4.4.42 Current Value of Parking Monitoring Mode

[Interface Function]

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Query the current value of the setting item of the parking monitoring mode on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamvalue

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=parking_mode: Obtain the current value of the parking monitoring mode.

[Return Parameter on the Device]

Parameter	Type	Description
value	int	The parameter value corresponding to the setting item parking_mode corresponds to the value in the index array in the parameter list: 0: off 1: timelapse 2: normrec (normal recording)

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "value" : 0
    }
}
```

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4.4.43 Current Value of Parking Monitoring Switch

[Interface Function]

Query the current value of the setting item of the parking monitoring switch on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamvalue

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=parking_monitor: Obtain the current value of parking monitoring switch.

[Return Parameter on the Device]

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Parameter	Type	Description
value	int	The parameter value corresponding to the setting item parking_monitor corresponds to the value in the index array in the parameter list: 1: on 0: off

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "value" : 0
    }
}
```

4.4.44 Current Value of Light Frequency

[Interface Function]

Query the current value of the setting item of the light frequency on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamvalue

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=light_fre: Obtain the current value of the light frequency.

[Return Parameter on the Device]

Parameter	Type	Description
value	int	The parameter value corresponding to the setting item light_fre corresponds to the value in the index array in the parameter list: 0: 50Hz 1: 60Hz

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
```

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```
        "value" : 0
    }
}
```

4.4.45 Current Value of Rear Mirror Switch

[Interface Function]

Query the current value of the setting item of the rear mirror switch on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamvalue

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=rear_mirror: Obtain the current value of the rear mirror switch.

[Return Parameter on the Device]

Parameter	Type	Description
value	int	The parameter value corresponding to the setting item rear_mirror corresponds to the value in the index array in the parameter list: 1: on 0: off

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "value" : 0
    }
}
```

4.4.46 Current Value of Wide Dynamic Switch

[Interface Function]

Query the current value of the setting item of the wide dynamic switch on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamvalue

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[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=wdr: Obtain the current value of the wide dynamic switch.

[Return Parameter on the Device]

Parameter	Type	Description
value	int	The parameter value corresponding to the setting item wdr corresponds to the value in the index array in the parameter list: 1: on 0: off

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "value" : 0
    }
}
```

4.4.47 Current Value of Screen Standby Duration

[Interface Function]

Query the current value of the setting item of the screen standby duration on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamvalue

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=screen_standby: Obtain the current value of the screen standby duration.

[Return Parameter on the Device]

Parameter	Type	Description
value	int	The parameter value corresponding to the setting item screen_standby corresponds to the value in the index array in the parameter list:

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Parameter	Type	Description
		0: off 1: 10s 2: 30s 3: 60s

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "value" : 0
    }
}
```

4.4.48 Current Value of Auto Poweroff Duration

[Interface Function]

Query the current value of the setting item of the auto poweroff duration on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamvalue

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=auto_poweroff: Obtain the current value of the auto poweroff duration.

[Return Parameter on the Device]

Parameter	Type	Description
value	int	The parameter value corresponding to the setting item auto_poweroff corresponds to the value in the index array in the parameter list: 0: off 1: 1H 2: 12H

[Example of Device Return after Success]

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```
{
    "result" : 0,
    "info" : {
        "value" : 0
    }
}
```

4.4.49 Current Value of ADAS Switch

[Interface Function]

Query the current value of the setting item of the ADAS switch on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamvalue

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=adas: Obtain the current value of the ADAS switch.

[Return Parameter on the Device]

Parameter	Type	Description
value	int	The parameter value corresponding to the setting item adas corresponds to the value in the index array in the parameter list: 1: on 0: off

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "value" : 0
    }
}
```

4.4.50 Current Value of Boot Sound Switch

[Interface Function]

Query the current value of the setting item of the boot sound switch on the device.

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[Request Interface on the APP]

http://192.168.169.1/app/getparamvalue

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=boot_sound: Obtain the current value of the ADAS switch.

[Return Parameter on the Device]

Parameter	Type	Description
value	int	The parameter value corresponding to the setting item boot_sound corresponds to the value in the index array in the parameter list: 1: on 0: off

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "value" : 0
    }
}
```

4.4.51 Current Value of Video Vertical Flip Switch

[Interface Function]

Query the current value of the setting item of the video vertical flip switch on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamvalue

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=video_filp: Obtain the current value of the video vertical flip switch.

[Return Parameter on the Device]

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Parameter	Type	Description
value	int	The parameter value corresponding to the setting item video_filp corresponds to the value in the index array in the parameter list: 1: on 0: off

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "value" : 0
    }
}
```

4.4.52 Current Value of Video Horizontal Mirror Switch

[Interface Function]

Query the current value of the setting item of the video horizontal mirror switch on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamvalue

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=video_mirror: Obtain the current value of the video horizontal mirror switch.

[Return Parameter on the Device]

Parameter	Type	Description
value	int	The parameter value corresponding to the setting item video_mirror corresponds to the value in the index array in the parameter list: 1: on 0: off

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
```

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```
        "value" : 0
    }
}
```

4.4.53 Current Value of Key Tone Switch

[Interface Function]

Query the current value of the setting item of the key tone switch on the device.

[Request Interface on the APP]

http://192.168.169.1/app/getparamvalue

[Request Parameter on the APP]

Parameter	Type	Description
param	String	?param=key_tone: Obtain the current value of the key tone switch.

[Return Parameter on the Device]

Parameter	Type	Description
value	int	The parameter value corresponding to the setting item key_tone corresponds to the value in the index array in the parameter list: 1: on 0: off

[Example of Device Return after Success]

```
{
    "result" : 0,
    "info" : {
        "value" : 0
    }
}
```

4.4.54 List of Recommended Common Setting Items

All common settings are queried and set based on the rules of the above interfaces, and the setting items are implemented and extended by the device, and the recommended device settings are as follows:

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Table 4-1 List of Recommended Common Setting Items

Setting Items	Description
Microphone switch	mic switch: 0-off (close) 1-on (open)
Time watermark switch	osd switch: 0-off (close) 1-on (open)
Logo watermark switch	logo_osd switch: 0-off (close) 1-on (open)
Recording resolution	rec resolution: 0-720P 1-1080P 2-1296P 3-2K
Recording file duration	rec split duration: 0-1MIN (1 minute) 1-2MIN (2 minutes)
Video encoding format	encodec: 0-h.264 1-h.265
Speaker volume	speaker: 0-off 1-low 2-middle 3-high 4-very high
Collision sensitivity	gsr_sensitivity: 0-high 1-low

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Setting Items	Description
	2-off
Recording status	rec switch: 0-off (stop recording) 1-on (start recording)
Language type	language: 0-en_US 1-zh_CN (Simplified Chinese) 2-zh_CHT (Traditional Chinese) 3-ja (Japanese) 4-ru (Russian) 5-th (Thai) 6-cran (Vietnamese)
Timelapse recording frame rate	timelapse_rate: 0-off 1-1fps 2-2fps 3-5fps
Parking monitoring recording duration	park_record_time: 0-off 1-2hour 2-24hour 3-48hour
Parking monitoring collision sensitivity	park_gsr_sensitivity: 0-high 1-low 2-off
Parking minotoring mode	parking_mode: 0-off 1-timelapse 2-normrec (normal recording)

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Setting Items	Description
Parking monitoring switch	parking_monitor switch: 0-off 1-on
Light frequency	light_fre: 0-50Hz 1-60Hz
Rear mirror switch	rear_mirror: 0-off 1-on
Wide dynamic switch	wdr switch: 0-off 1-on
Voice control switch	voice_control switch: 0-off 1-on
Screen standby duration	screen_standby: 0-off 1-10s 1-30s 1-60s
Auto poweroff duration	auto_poweroff: 0-off 1-1H 2-12H
ADAS switch	adas switch: 0-off 1-on
Boot sound switch	boot_sound switch: 0-off 1-on

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Setting Items	Description
Video vertical flip switch	video_filp switch: 0-off 1-on
Video horizontal mirror switch	video_mirror switch: 0-off 1-on
Key tone switch	key_tone switch: 0-off 1-on

CAUTION
The parameter flags of all setting items need to be in lowercase.

4.5 Push Message

The push message is a state change message sent to the APP by the device through TCP socket (Note: the persistent connection is used).

4.5.1 Message of Recording Status Change

[Interface Function]

Notify the recording state change (Starting or stopping of recording) of the device through other operations.

[Request Interface on the APP]

None.

[Request Parameter on the APP]

None.

[Return Interface on the Device]

IP	Port Number
192.168.169.1	5000

[Return Parameter on the Device]

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Parameter	Type	Description
msgid	String	Message ID. It indicates an event type.
info	Json Object	Message content object corresponding to msgid.
value	int	The parameter value corresponding to the setting item msgid corresponds to the value in the index array in the parameter list: 0: start 1: stop
time	long	Transmission time of the message, Unix timestamp (Unix createtimestamp), the unit is s.

[Example of Device Return after Success]

```
{
    "msgid" : "rec",
    "info" : {
        "value" : 1
    },
    "time" : 1593422089
}
```

4.5.2 Message of Microphone Status Change

[Interface Function]

Notify the microphone state change (Opening or closing of microphone) of the device through other operations.

[Request Interface on the APP]

None.

[Request Parameter on the APP]

None.

[Return Interface on the Device]

IP	Port Number
192.168.169.1	5000

[Return Parameter on the Device]

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Parameter	Type	Description
msgid	String	Message ID. It indicates an event type.
info	Json Object	Message content object corresponding to msgid.
value	int	The parameter value corresponding to the setting item msgid corresponds to the value in the index array in the parameter list: 0: off 1: on
time	long	Transmission time of the message, Unix timestamp (Unix createtimestamp), the unit is s.

[Example of Device Return after Success]

```
{
    "msgid" : "mic",
    "info" : {
        "value" : 1
    },
    "time" : 1593422089
}
```

4.5.3 Message of Battery Status Change

[Interface Function]

It is used for reporting the change of battery state on the device. For example: charging state change, battery state change.

[Request Interface on the APP]

None.

[Request Parameter on the APP]

None.

[Return Interface on the Device]

IP	Port Number
192.168.169.1	5000

[Return Parameter on the Device]

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Parameter	Type	Description
msgid	String	Message ID. It indicates an event type.
info	Json Object	Message content object corresponding to msgid.
capacity	int	0-100. Indicate the percentage value of the current battery capacity.
charge	int	0: Not charged; 1: Charging
time	long	Transmission time of the message, Unix timestamp (Unix createtimestamp), the unit is s.

[Example of Device Return after Success]

```
{
    "msgid" : "battery",
    "info" : {
        "capacity" : 0,
        "change" : 1
    },
    "time" : 1593422089
}
```

4.5.4 Message of Storage Card Status Change

[Interface Function]

It is used for reporting the change of SD card or TF card on the device. For example, the reported notice has the status of inserted, not inserted, damaged, unknown exception, etc.

[Request Interface on the APP]

None.

[Request Parameter on the APP]

None.

[Return Interface on the Device]

IP	Port Number
192.168.169.1	5000

[Return Parameter on the Device]

Parameter	Type	Description
msgid	String	Message ID. It indicates an event type.

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Parameter	Type	Description
info	Json Object	Message content object corresponding to msgid.
status	int	0: SD/TF card is normal 1: SD/TF card is unformatted 2: SD/TF card is pulled out ... 99: Unknown exception
time	long	Transmission time of the message, Unix timestamp (Unix createtimestamp), the unit is s.

[Example of Device Return after Success]

```
{
    "msgid" : "sd",
    "info" : {
        "status" : 0
    },
    "time" : 1593422089
}
```

4.5.5 Message of File Deletion Status Change

[Interface Function]

It is used for reporting that the video or picture file on the device is deleted. If the deletion is caused by operations on the APP, it is recommended not to report such messages.

[Request Interface on the APP]

None.

[Request Parameter on the APP]

None.

[Return Interface on the Device]

IP	Port Number
192.168.169.1	5000

[Return Parameter on the Device]

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Parameter	Type	Description
msgid	String	Message ID. It indicates an event type.
info	Json Object	Message content object corresponding to msgid.
name	String	Path name corresponding to the deleted file.
type	int	File type. 1: Picture file; 2: Video file
time	long	Transmission time of the message, Unix timestamp (Unix createtimestamp), the unit is s.

[Example of Device Return after Success]

```
{
    "msgid" : "file_del",
    "info" : {
        "name" : "/mnt/sdcard/VIDEO_F/20201120190300_f.ts",
        "type" : 2
    },
    "time" : 1593422089
}
```

4.5.6 Message of File Addition Status Change (Option)

[Interface Function]

The message after adding a file on the device is reported. When the device adds a file, it needs to be reported if the APP wants to know and refresh the file list at the first time.

[Request Interface on the APP]

None.

[Request Parameter on the APP]

None.

[Return Interface on the Device]

IP	Port Number
192.168.169.1	5000

[Return Parameter on the Device]

Parameter	Type	Description
msgid	String	Message ID. It indicates an event type.

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Parameter	Type	Description
info	Json Object	Message content object corresponding to msgid.
name	String	Path name corresponding to the added file.
type	int	File type. 1: Picture file; 2: Video file
time	long	Transmission time of the message, Unix timestamp (Unix createtimestamp), the unit is s.

[Example of Device Return after Success]

```
{
    "msgid" : "file_add",
    "info" : {
        "name" : "/mnt/sdcard/VIDEO_F/20201120190300_f.ts",
        "type" : 2
    },
    "time" : 1593422089
}
```

4.5.7 Message of File Locking Status Change

[Interface Function]

When the current recording file on the device is locked through other operations (other than on the APP), this message is reported to the APP.

[Request Interface on the APP]

None.

[Request Parameter on the APP]

None.

[Return Interface on the Device]

IP	Port Number
192.168.169.1	5000

[Return Parameter on the Device]

Parameter	Type	Description
msgid	String	Message ID. It indicates an event type.
info	Json Object	Message content object corresponding to msgid.

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Parameter	Type	Description
value	int	Current corresponding value. It should correspond to the value array of the msgid. 0: Stop locking; 1: Start locking
time	long	Transmission time of the message, Unix timestamp (Unix createtimestamp), the unit is s.

[Example of Device Return after Success]

```
{
    "msgid" : "rec_lock",
    "info" : {
        "value" : 1
    },
    "time" : 1593422089
}
```

4.5.8 Message of Real-Time GPS Information Status Change

[Interface Function]

It is used to report the current motion state of GPS, such as speed, longitude/latitude, Gsensor, altitude, and so on.

[Request Interface on the APP]

None.

[Request Parameter on the APP]

None.

[Return Interface on the Device]

IP	Port Number
192.168.169.1	5000

[Return Parameter on the Device]

Parameter	Type	Description
msgid	String	Message ID. It indicates an event type.
info	Json Object	Message content object corresponding to msgid.
value	String	Current corresponding GPS information. It needs to meet the parameter format of the protocol.

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Parameter	Type	Description
time	long	Transmission time of the message, Unix timestamp (Unix createtimestamp), the unit is s.

[Parameter Format]

Parameters	Instances
Date occurred	2023/07/06 (Data format: yyyy/MM/dd)
Time occurred	16:25:59 (Time format: HH:mm:ss)
Latitude	N:22.524790 (N: North latitude, S: South latitude, unit: degree)
Longitude	E:113.935379 (E: East longitude, W: West longitude, unit: degree)
Speed	80.00
Unit of speed	km/h (Other unit needs to be converted to km/h)
Heading	271.80 (Unit: degree, parameter range: [0~360].)
Altitude	500.00 (Unit: m, parameter range: [-9999~9999].)
Number of satellites currently searched	17 (Unit: pcs. It is used to display the signal intensity of GPS.)
X-axis data of Gsensor	-0.004 (Unit: G)
Y-axis data of Gsensor	-0.137 (Unit: G)
Z-axis data of Gsensor	+0.054 (Unit: G)

[Example of Content Data]

- Valid data

```
2023/07/06 16:25:59 N:22.524790 E:113.935379 80.00 km/h 271.80 500.00 17 x:-0.004
    y:-0.137 z:+0.054
```

- Data with invalid fields (The current environment does not support, such as in the tunnel)

```
2023/07/06 16:25:59 NA NA NA km/h NA NA NA x:-0.004 y:-0.137 z:+0.054
```

- Not supported fields by the current GPS module

```
2023/07/06 16:25:59 N:22.524790 E:113.935379 80.00 km/h - - - x:-0.004 y:-0.137 z
    :+0.054
```

[Example of Device Return after Success]

```
{
    "msgid" : "gps",
    "info" : {
```

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```
        "value" : "2023/07/06 16:25:59 N:22.524790 E:113.935379 80.00 km/h 271.80
    500.00 17 x:-0.004 y:-0.137 z:+0.054"
    },
    "time" : 1688631959
}
```

CAUTION

To maintain compatibility, the GPS data must contain the fields listed in the above table. Fields that are not supported or are not available are filled according to the following rule.
'NA' indicates the field value of this data is invalid, so there is no value; if the device does not support the field, you can use '-', and each field needs to be separated by a space.

4.5.9 Message of GPS Information Status Change

[Interface Function]

The device periodically reports the current number of GPS satellites through a TCP persistent connection, which will be used after the APP enters the setting interface to view the current GPS satellite number and signal on the device.

[Request Interface on the APP]

None.

[Request Parameter on the APP]

None.

[Return Interface on the Device]

IP	Port Number
192.168.169.1	5000

[Return Parameter on the Device]

Parameter	Type	Description
msgid	String	Message ID. It indicates an event type.
info	Json Object	Message content object corresponding to msgid.
value	String	Number of satellites currently searched and corresponding signal values. It is suggested that the number of satellites should be fixed on the device, and the signal value of the satellites without signal should be set to 0.

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Parameter	Type	Description
time	long	Transmission time of the message, Unix timestamp (Unix createtimestamp), the unit is s.

[Example of Device Return after Success]

```
{
    "msgid" : "gps_satellite",
    "info" : {
        "value" : "10,12,5,3,6,8,6,0,0,0"
    },
    "time" : 1688631959
}
```

4.5.10 Message of Camera Plugging Status Change

[Interface Function]

It is used for reporting the current state of camera plugging/unplugging. In the process of using the APP, when the camera of the device is plugged and unplugged, the message is reported to the APP.

[Request Interface on the APP]

None.

[Request Parameter on the APP]

None.

[Return Interface on the Device]

IP	Port Number
192.168.169.1	5000

[Return Parameter on the Device]

Parameter	Type	Description
msgid	String	Message ID. It indicates an event type.
info	Json Object	Message content object corresponding to msgid.
action	int	0: plugout; 1: plugin
curcamid	int	ID of the camera currently viewed (the camera currently previewed after changing). It starts from 0.
camnum	int	Number of the current cameras (after changing).

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Parameter	Type	Description
time	long	Transmission time of the message, Unix timestamp (Unix createtimestamp), the unit is s.

[Example of Device Return after Success]

```
{
    "msgid" : "cam_plugin",
    "info" : {
        "action" : 1,
        "curcamid" : 1,
        "camnum" : 2,
    },
    "time" : 1593422089
}
```

4.6 Control Real-Time Stream

Table 4-2 Control Real-Time Stream

Real-Time Stream	Note
Description	Transmission control for RTSP real-time stream.
Interface	rtsp://192.168.169.1:554 or rtsp://192.168.169.1:8554.
Transfer layer	UDP or TCP (It is best to use TCP). You can set **RTSP over UDP** or **RTSP over TCP** according to the transport parameter value in the http://192.168.169.1/app/getmediainfo request.
RTSP version	RTSP/1.0
Methods supported by RTSP	OPTIONS, DESCRIBE, SETUP, PLAY, TEARDOWN
Prerequisite for opening RTSP server on the device	1. Send the http://192.168.169.1/app/enterrecorder request (option); 2. Send heartbeat regularly (such as every 5 seconds): http://192.168.169.1/app/getparamvalue?param=rec (option); 3. Send the http://192.168.169.1/app/getmediainfo request (option).

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Real-Time Stream	Note
Media encoding format supported by the pushflow on the device	Video: H.264/H.265; Audio: AAC

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